Leveraging Inference-Time Compute for Diffusion Models via Global Scheduling of Denoising Trajectories

Anonymous Author(s)
Affiliation
Address
email

Abstract

Diffusion models generate samples by traversing a denoising trajectory, a sequence of noise-reduction steps that progressively transforms random noise into data of the target distribution. A practical question at inference time is how to use a given compute budget to generate higher-quality samples without retraining. Standard approaches allocate compute uniformly across the denoising steps, but this is suboptimal: early steps that determine global structure often benefit more from additional refinement than late steps that refine local details. We formalize this as a budget allocation problem: at each timestep t, evaluate K_t noise candidates and keep the best (under a fixed quality scorer, the *verifier*); choose $\{K_t\}_{t=1}^T$ to maximize the expected verifier score of the final sample subject to $\sum_t K_t = B$ and $K_t \geq 1$. We develop a theoretical model for this problem and establish two optimality results. First, the optimal offline allocation has a closed-form water-filling rule: each step's share grows with its sensitivity to noise refinement. Second, we prove an $\Omega(T)$ worst-case regret lower bound against any adaptive policy, motivating an offline-plus-online design rather than pure online adaptation. We instantiate this analysis as GAINS (Global Adaptive Inference-time Noise Scheduling), an algorithm that pairs offline sensitivity profiling with online feedback control. On Stable Diffusion, EDM, and flow-based models, GAINS matches the quality of uniform allocation using 20–50% fewer function evaluations, and exceeds it at equal budget.

1 Introduction

Diffusion models generate data by following a *denoising trajectory*: starting from random noise, the model applies a sequence of learned noise-reduction steps until a sample from the target distribution emerges [Ho et al., 2020, Song and Ermon, 2019, Song et al., 2021b]. Generating one sample takes T such denoising steps, each costing one forward pass through the network; we call each pass a *function evaluation* (NFE), so standard sampling costs T NFE. Recent work shows that spending more than T NFE per sample, a regime known as *inference-time scaling*, can improve sample quality without retraining [Ma et al., 2025, Ramesh and Mardani, 2025]. This raises a resource-allocation question: how should the extra NFE be distributed across the T denoising steps? The extra budget can enter in two ways: (i) *lengthening the trajectory* by increasing T, which empirically saturates [Ma et al., 2025], and (ii) *searching within each timestep* by evaluating multiple noise candidates and keeping the one that scores highest on a quality metric. Once the trajectory is long enough that (i) gives diminishing returns, additional NFE is best spent on (ii); we focus on this within-step search regime, which is the lever exploited by recent inference-time methods.

Within-step search is implemented as a local operation on the noise variable that drives each step. Concretely, the model draws several candidate noise variables, scores each via a *verifier* (a scalar quality function, e.g., a CLIP-style image-text alignment score), and keeps the highest-scoring one, obtaining a better sample at the cost of additional NFE. Existing methods instantiate this local noise search in various forms (greedy, zero-order, and particle-based search; Ramesh and Mardani 2025, Tang et al. 2024, Kim et al. 2025).

These methods all enrich the local operator within a single step, but distribute the search budget uniformly across denoising steps or fix it to a hand-chosen window (e.g., only the initial noise, or a predetermined middle segment). Yet replacing a noise draw with a better candidate empirically improves quality far more at some timesteps than at others; we call this per-step responsiveness the *sensitivity* of step t. This leaves a complementary question: given a fixed NFE budget, *where along the trajectory* should search concentrate?

We cast *noise trajectory search* as a budget-allocation problem: choose per-step counts $K_t \in \mathbb{Z}_{\geq 1}$ that maximize the expected verifier score subject to $\sum_t K_t = B$. The problem decomposes into two levels coupled only through K_t: a *local operator* searches noise candidates within one timestep, and a *global scheduler* splits B across the T timesteps. The local level is well studied; the global level, which allocates differently across timesteps with unequal sensitivity, is the gap we address. This decoupling lets any existing local operator (greedy, zero-order, particle-based) compose with any global schedule; standard sampling, best-of-N, ϵ-greedy, and particle/tree-based methods become special cases with hard-coded or uniform global policies.

Building on this formulation, we propose **GAINS** (**G**lobal **A**daptive **I**nference-time **N**oise **S**cheduling), a global scheduling algorithm that combines *offline profiling* with *online control*. Offline, GAINS profiles verifier sensitivity across timesteps to obtain a coarse compute allocation specific to the model and dataset. Online, GAINS adjusts the search at each step using the verifier score improvement and candidate variance observed so far, stopping early on unproductive steps and reallocating saved budget to later ones. The controller operates causally (no access to future timesteps) and respects a strict NFE budget.

Contributions.

1. We introduce a *two-level view* of noise trajectory search that separates local noise refinement from global timestep scheduling, covering both stochastic and deterministic samplers and recovering existing inference-time methods as special-case scheduling policies (Section 3).
2. We develop GAINS, a *global scheduling algorithm* that integrates offline timestep profiling with online feedback control, enforcing an exact NFE budget (Section 4).
3. We give a *theoretical model* for the noise trajectory search problem with two optimality results: a closed-form water-filling characterization of the optimal offline allocation (Theorem 6), and a matching worst-case regret lower bound of order T (Lemma 7) for a sequential-allocation game into which the diffusion problem embeds. The lower bound shows that no online-only policy avoids regret of order T in the worst case, motivating the hybrid design.

Empirically, on text-to-image (Stable Diffusion), class-conditional (EDM [Karras et al., 2022]), and flow-based generation, GAINS matches or exceeds uniform-allocation benchmarks at fixed NFE budget, recovering the same Brightness and Compressibility verifier scores with 20–50% fewer NFE, and adds $+0.005$ to $+0.04$ at equal budget; the gains hold across local search operators, prompt sets, and model families (Section 5).

1.1 Related Literature

Inference-time noise search. A growing body of work improves diffusion sample quality by refining noise variables at generation time, paralleling LLM inference-time scaling [Snell et al., 2024, Brown et al., 2024]. Early approaches select among multiple full trajectories [Ma et al., 2025] or optimize noise via gradient- or zero-order updates per timestep [Tang et al., 2024, Qi et al., 2024, Zhou et al., 2025]. More recent methods introduce richer local operators along three threads: greedy or two-phase search at each step [Ramesh and Mardani, 2025, Kim et al., 2025, Zhang et al., 2025b, Lee et al., 2025, Yu et al., 2025], particle- and tree-based search with non-uniform particle budgets across stages [Singhal et al., 2025, Yoon et al., 2025, Zhang et al., 2025c], and amortized or learned per-step compute [Eyring et al., 2025, Zhang et al., 2025a, Ye et al., 2025, Yeh et al., 2025, Sundaresha et al., 2025]. The third thread modifies per-step effort but primarily adapts the base sampler rather than scheduling additional search compute. These methods improve sample quality by designing richer *local* search operators applied within each timestep. The complementary question of *how much* budget each timestep should receive is typically resolved by uniform allocation or a hand-tuned schedule (e.g., particle counts fixed in advance in Singhal et al. [2025], Yoon et al. [2025]); we build on these local operators and address the global allocation question directly. GAINS estimates a per-step sensitivity profile from a small calibration set and adapts allocation online based on observed verifier-score gains, distributing a fixed NFE budget non-uniformly across the trajectory.

Connections to bandits and resource allocation. Each component of the problem has a classical counterpart. The local operator (evaluate K noise candidates, keep the best) is a fixed-budget best-arm identification problem: split a sample budget across arms with unknown means and return the empirical best [Chen et al., 2000, Kaufmann et al., 2016, Lattimore and Szepesvári, 2020]. The global scheduler is a sequential budget-allocation problem across the T steps [Ibaraki and Katoh, 1988, Boyd and Vandenberghe, 2004]. Our contribution to this perspective is a result specific to the diffusion setting (Theorem 5) that decomposes the per-step expected gain into a sensitivity factor and a sample-size factor; this decomposition reduces the across-timestep allocation problem to a closed-form water-filling solution and gives a matching regret lower bound of order T.

Theoretical foundations of diffusion models. Several recent theoretical results provide useful context. Gao et al. [2024] develop a continuous-time RL framework for reward-directed diffusion and show that the optimal exploration policy has time-varying variance, a structure consistent with the location-scale model we adopt. Tang and Zhao [2024] characterize a per-timestep error sensitivity for score-estimation propagation, conceptually related to our per-step sensitivity but capturing estimation rather than verifier-score sensitivity; Chen et al. [2023] establish polynomial convergence of score-based samplers under minimal data assumptions, foundational for the SDE discretization we use in Section 4.3. Our work complements these analyses by focusing on the budget allocation problem: we show that per-step gains decompose into a sensitivity factor and a sample-size effect, and use this structure to characterize optimal scheduling.

2 Preliminaries and Problem Setting

We fix notation for diffusion-model sampling, define the verifier and NFE budget, and state the noise trajectory search problem.

Diffusion models. A diffusion model generates a sample in $\mathbb{R}^d$ over T discrete denoising steps, indexed $t = T, T-1, \ldots, 1$. Write $x_t \in \mathbb{R}^d$ for the latent at step t (x_T is the initial Gaussian noise, x_0 the final sample). The Denoising Diffusion Probabilistic Modeling (DDPM) formulation [Ho et al., 2020] couples a forward process that perturbs data with Gaussian noise and a backward process that converts noise back to data. Setting $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$ for a fixed noise schedule $\beta_t \in (0, 1)$, samples emerge by iterating

$$x_{t-1} = \tfrac{1}{\sqrt{1-\beta_t}}\Big(x_t - \tfrac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\, \varepsilon_\theta(x_t, t)\Big) + \tilde{\sigma}_t\, \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, I_d), \tag{1}$$

from $x_T \sim \mathcal{N}(0, I_d)$ to x_0, where ε_θ is a learned noise predictor, $\tilde{\sigma}_t$ is the (fixed) reverse-process noise scale, and $\varepsilon_t \in \mathbb{R}^d$ is the per-step injected Gaussian noise (with I_d the $d \times d$ identity). Score-based [Song and Ermon, 2019], flow-based [Lipman et al., 2023, Liu et al., 2023], and accelerated samplers [Song et al., 2021a, Lu et al., 2022] share the same structure: a T-step backward trajectory driven by per-step stochastic noise ε_t.

Inference-time noise search.

Definition 1 (Verifier)**.** Let $\mathcal{X} \subseteq \mathbb{R}^d$ denote the sample space (e.g., images or latents) and $\mathcal{C}$ the space of conditioning signals (e.g., text prompts or class labels). A *verifier* $v : \mathcal{X} \times \mathcal{C} \to \mathbb{R}$ scores a sample $x \in \mathcal{X}$ against signal $c \in \mathcal{C}$ (analogous to outcome/process verifiers for LLMs [Cobbe et al., 2021, Lightman et al., 2023]). It is fixed during search and treated as a black box; results hold for any bounded, query-efficient v.

Definition 2 (NFE budget)**.** A single *function evaluation* (NFE) is one forward pass through the denoising network, the dominant cost in diffusion sampling. Standard sampling consumes exactly T

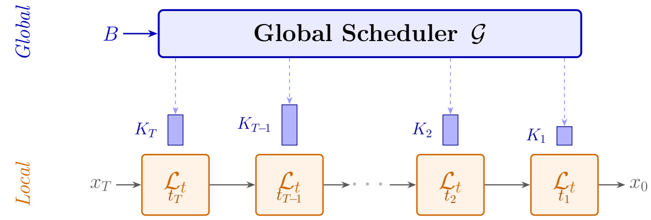

Figure 1: Two-level view of noise trajectory search. The *global scheduler* $\mathcal{G}$ (top) receives budget B and distributes per-step budgets $\{K_t\}$ across the trajectory. At each timestep, the *local search operator* $\mathcal{L}_t$ (bottom) evaluates K_t noise candidates and selects the best. Bars depict the non-uniform allocation: timesteps where replacing the noise candidate yields larger quality gains receive bigger budgets.

NFE; noise trajectory search spends the remaining $B - T$ NFE on local refinement, where $B \geq T$ is the total budget.

Definition 3 (Noise trajectory search problem)**.** Given a pretrained diffusion model with one-step transition $F_\theta : \mathbb{R}^d \times \{1, \dots, T\} \times \mathbb{R}^d \to \mathbb{R}^d$ that maps (x_t, t, ε_t) to x_{t-1} (the abstraction of Equation (1), formalized in Equation (2)), a conditioning signal $c \in \mathcal{C}$, a verifier $v(\cdot, c)$ (Definition 1), and a total budget of B function evaluations (Definition 2), find a sequence $\{\varepsilon_t^*\}_{t=1}^T$ that maximizes the verifier score of the final sample x_0 subject to total NFE $\leq B$.

The difficulty is sequential: ε_t shapes x_{t-1}, which determines which $\varepsilon_{t'}$ are effective at later $t' < t$. This motivates the two-level decomposition of Section 3.

3 A Two-Level Framework for Noise Trajectory Search

We split noise trajectory search into two components on top of a model-agnostic transition abstraction (Figure 1): a *local search operator* $\mathcal{L}_t$ that explores noise candidates at a single timestep, and a *global scheduler* $\mathcal{G}$ that decides how to spend the budget B across timesteps. The two components interact only through the per-step budget K_t, so any local operator can be paired with any global scheduler, and we can study global allocation while reusing existing local operators.

General transition abstraction. Every diffusion-family model generates a sample by traversing T discrete timesteps $t_T > t_{T-1} > \cdots > t_1$. At each step the sampler applies

$$x_{t-1} = F_\theta(x_t, t, c, \varepsilon_t), \tag{2}$$

where x_t is the latent at timestep t, c is the conditioning signal, and ε_t is a noise variable injecting stochasticity. The final step yields x_0, which the verifier $v(x_0, c)$ scores. The transition is deliberately model-agnostic: for stochastic samplers (DDPM, SDE) ε_t is the injected Gaussian noise; for deterministic samplers (flow-matching ODE solvers [Lipman et al., 2023]) stochasticity can be injected by replacing a deterministic ODE step with an SDE step of matched marginal [Singh and Fischer, 2024, Kim et al., 2025].

Local search operator. At timestep t, given (x_t, c, v, F_θ) and a per-step budget K_t, the local operator samples K_t noise candidates $\{\varepsilon_t^{(k)}\}_{k=1}^{K_t}$ and applies the base sampler F_θ (one timestep advance) to each, then maps each resulting latent to a clean-data prediction $\hat{x}_0^{(k)}$ via the standard one-step denoiser D_θ [Song et al., 2021b] (the denoiser predicts the final clean sample $\hat{x}_0$ from any intermediate noisy latent, distinct from F_θ):

$$x_{t-1}^{(k)} = F_\theta(x_t, t, c, \varepsilon_t^{(k)}), \qquad \hat{x}_0^{(k)} = D_\theta(x_{t-1}^{(k)}, t-1, c). \tag{3}$$

We write $\mathcal{L}_t : (x_t, c, v, F_\theta, K_t) \to x_{t-1}$ for the local operator. The operator selects the candidate that maximizes the verifier score $s^{(k)} = v(\hat{x}_0^{(k)}, c)$. Each candidate costs one NFE, so the operator consumes K_t NFE in total. Many existing search procedures (zero-order perturbation [Tang et al., 2024], ϵ-greedy exploration [Ramesh and Mardani, 2025], particle-based refinement [Kim et al., 2025]) instantiate this operator; for our purposes $\mathcal{L}_t$ is a black-box mapping from x_t to x_{t-1} whose only knob is the per-step budget K_t.

Global scheduler. The local operator decides *how* to search at a given step but not *where along the trajectory* to invest computation. A global scheduler $\mathcal{G} : \left(x_T, c, v, F_\theta, B, \{\mathcal{L}_t\}_{t=1}^T\right) \longrightarrow \left(x_0, \{x_t\}_{t=1}^T\right)$ walks timesteps from T to 1, and at each step picks a per-step budget $K_t \geq 1$ subject to $\sum_t K_t \leq B$: setting $K_t = 1$ applies a single base step, while $K_t > 1$ invokes $\mathcal{L}_t$ to evaluate K_t noise candidates. This allocation must be made *online*: at step t we cannot predict how effective search will be at later timesteps, yet computation spent now is lost forever.

Existing methods as special cases. This view recovers several prior strategies as special cases. Plain diffusion sampling: $K_t = 1$ for all t. Best-of-N random search: $K_T = N$, $K_{t<T} = 1$. Noise-trajectory and tree-style search: specific $\mathcal{L}_t$ paired with a hard-coded $\{K_t\}$. For richer action spaces (multiple verifier types, backtracking), Appendix A extends the formulation to a Markov decision process.

4 The GAINS Algorithm and Theoretical Analysis

GAINS (**G**lobal **A**daptive **I**nference-time **N**oise **S**cheduling) operates at two granularities: offline coarse allocation (Section 4.1) sets per-timestep budgets from a small calibration run; online control (Section 4.2) adapts them per prompt. The theory (Section 4.3) shows the per-step gain factorizes into a sensitivity factor times a sample-size factor, giving a closed-form water-filling allocation; an $\Omega(T)$ worst-case regret lower bound motivates online adaptation. Extensions to ϵ-greedy/local-perturbation operators appear in Appendix C.

4.1 Offline Coarse Time Scheduling

Offline scheduling decides, before any test-time prompt arrives, how many local-search iterations to spend at each timestep. We consider the common setting where the sampler keeps a single latent x_t and a single noise sample ε_t at each timestep but performs multiple local search iterations before committing to x_{t-1}. The global scheduling problem then reduces to per-timestep decisions: at each t, given the search history and remaining budget, should we continue local search or stop and move on?

The difficulty is that, when deciding at t, we know neither (1) the additional improvement that further search at t would bring nor (2) the improvement we forgo by spending the same NFE at later timesteps. Offline profiling addresses (1) and (2) jointly by estimating, for each timestep, how responsive the verifier score is to local search. For each model and dataset, we run a one-time profiling pass on a small calibration set: noise search executes with a uniform iteration count at every timestep, logging verifier scores after each iteration. From these logs we compute, for each t, the average per-iteration score gain and its variability across examples; aggregating reveals which segments of the trajectory respond most to local search. The pattern is strongly model-dependent: for Stable Diffusion, gains concentrate on early timesteps; for EDM they cluster in a contiguous middle segment (Appendix D). We partition timesteps into a *high-value* region and a *low-value* region, allocate a larger share of B to the high-value region, and derive a coarse per-step budget $\{K_t\}$ (more iterations in high-value steps, fewer in low-value steps). The resulting schedule is tailored to the model and dataset but fixed across prompts; it captures *where on the trajectory search matters most*.

4.2 Online Fine-Grained Scheduling

Online scheduling refines, at test time, how the offline budget $\{K_t\}$ is spent on each individual prompt. At timestep t, local search runs up to K_t iterations, but the controller can stop early or, within a small window, shift spending across timesteps. We distinguish two scores: $\hat{s}_t^{(j)}$, the raw verifier score of the candidate produced in iteration j, and $s_t^{(j)} := \max(s_t^{(j-1)}, \hat{s}_t^{(j)})$, the running best after iteration j (with $s_t^{(0)} := \hat{s}_t^{(1)}$ from the first candidate). The controller monitors two statistics: the *score gain* $g_t^{(j)} := s_t^{(j)} - s_t^{(j-1)} \geq 0$ (which is 0 exactly when iteration j failed to improve), and the *score variance* across the candidates seen at step t,

$$\mathrm{Var}_t^{(j)} := \mathrm{Var}_{i \leq j}\left(\hat{s}_t^{(i)}\right). \tag{4}$$

The two statistics target complementary questions. The score gain $g_t^{(j)}$ asks whether the current iteration is *close to its local-search optimum*: when recent $g_t^{(j)}$ approach zero, the running best has

Algorithm 1 GAINS: Global Adaptive Inference-time Noise Scheduling

Input: timesteps $\{t_T, \dots, t_1\}$, total budget B, offline allocation $\{K_t\}$ ($\sum_t K_t = B$), slack δ, gain window W_g, threshold coefs (β_g, β_σ), black-box $\textsc{LocalIter}(t)$ that returns one raw candidate score $\hat{s}$ and within-step variance σ.
Output: realized per-step iterations $\{\widehat{K}_t\}$.
$R \leftarrow B$; $\mathcal{H}_g, \mathcal{H}_\sigma \leftarrow \emptyset$ (history).
for $t = T, T-1, \dots, 1$ **do**
  $K \leftarrow \min(K_t, R)$; $K_{\max} \leftarrow \min(K + \delta, R)$; $j_{\text{watch}} \leftarrow \max(2, K - \delta)$; $\mathcal{H} \leftarrow \emptyset$; $\widehat{K}_t \leftarrow 0$.
  $(\hat{s}^{(1)}, \sigma^{(1)}) \leftarrow \textsc{LocalIter}(t)$; $s^{(1)} \leftarrow \hat{s}^{(1)}$; $\widehat{K}_t \leftarrow 1$; $R \leftarrow R - 1$.
  for $j = 2, \dots, K_{\max}$ **do**
    $(\hat{s}^{(j)}, \sigma^{(j)}) \leftarrow \textsc{LocalIter}(t)$
    $s^{(j)} \leftarrow \max(s^{(j-1)}, \hat{s}^{(j)})$ {running best (revert-on-negative)}
    push $g^{(j)} \leftarrow s^{(j)} - s^{(j-1)} \geq 0$ into $\mathcal{H}$ (keep last W_g); $\widetilde{g}^{(j)} \leftarrow \mathrm{Mean}(\mathcal{H})$; $\widehat{K}_t \leftarrow \widehat{K}_t + 1$; $R \leftarrow R - 1$.
    if $R = 0$ **then**
      break
    end if
    if $j \geq j_{\text{watch}}$ **and** $|\mathcal{H}_g| > 0$ **and** $|\mathcal{H}_\sigma| > 0$ **and** $\widetilde{g}^{(j)} < \beta_g \mathrm{Mean}(\mathcal{H}_g)$ **and** $\sigma^{(j)} < \beta_\sigma \mathrm{Mean}(\mathcal{H}_\sigma)$ **then**
      break
    end if
  end for
  append $\mathrm{Mean}(\mathcal{H})$ to $\mathcal{H}_g$; append $\sigma^{(\widehat{K}_t)}$ to $\mathcal{H}_\sigma$.
end for
return $\{\widehat{K}_t\}$.

converged and additional NFE buys little. The score variance $\mathrm{Var}_t^{(j)}$ asks whether this timestep is *important*: by Theorem 5(ii), within-step variance equals σ_t^2 to leading order, so $\mathrm{Var}_t^{(j)}$ estimates the per-step sensitivity driving every gain in Section 4.3. Stopping early is justified only when *both* are small: a small gain alone can mark a high-sensitivity step waiting for a lucky draw, and small variance alone a low-sensitivity step whose iterations still nudge the running best upward; the conjunction prevents premature stopping in either regime (Figure 2). The *revert-on-negative* construction (running max) makes $s_t^{(j)}$ monotone, so any iteration is at worst a no-op. To match realized compute exactly to B, over-/under-spending from adaptive stopping in the high-value region is offset against low-value steps. Algorithm 1 gives the procedure.

4.3 Theoretical Foundations

Scope. The main-text results treat the local operator as **random search** (K i.i.d. Gaussian candidates per timestep), giving the cleanest factorization $\sigma_t \cdot a_K$; the same form extends to ϵ-greedy and local perturbation with operator-specific $\phi_K^{\mathcal{L}}$ replacing a_K in Appendix C. We justify GAINS in three steps. (1) A Taylor expansion of the verifier score around the noiseless Euler point gives $S_t = \mu_t + \sigma_t Z_t + O(g_t^2 h)$ with $Z_t \sim \mathcal{N}(0,1)$, where σ_t is the per-step *sensitivity* that governs how much one extra noise candidate is worth, and the expected K-candidate gain is $\sigma_t a_K$ to leading order (Theorem 5). (2) When σ_t depends only on t, the optimal allocation has closed-form water-filling structure (Theorem 6). (3) When $\sigma_t = \sigma_t(c, x_t)$ varies per prompt, no adaptive policy avoids worst-case regret of order T (Lemma 7; operator-agnostic). On non-adversarial prompt distributions, however, a positive Jensen gap (Remark 8) leaves per-prompt online control valuable.[1]

SDE setup. We treat the DDPM update Equation (1) as one Euler-Maruyama step [Song et al., 2021b] of a reverse-time SDE:

$$X_{t-h} = \bar{X}_{t-h} + g_t \sqrt{h}\, \xi, \quad \bar{X}_{t-h} := X_t + b_t(X_t, c)\, h, \quad \xi \sim \mathcal{N}(0, I), \tag{5}$$

[1]Notation: g_t (diffusion coefficient, below) differs from $g_t^{(j)}$ (per-iteration score gain, Section 4.2); σ_t (sensitivity, below) differs from σ (within-step empirical std, Algorithm 1).

with h the integration step, b_t the drift, g_t the diffusion coefficient (both schedule-fixed), $\bar{X}_{t-h}$ the noiseless Euler point, and ξ the per-step noise the local operator searches over. Let D_θ be the one-step denoiser [Song et al., 2021b], $\Psi_t(x;c) := v(D_\theta(x, t-h, c); c)$ the verifier score on the one-step denoising, and $S_t := \Psi_t(X_{t-h}; c)$. $\nabla\Psi_t$ denotes the gradient in the first argument at $\bar{X}_{t-h}$.

Assumption 4 (Verifier smoothness)**.** $\Psi_t(\cdot\,; c) \in C^2(\mathbb{R}^d)$, and there exists $L_t < \infty$ such that $\|\nabla^2\Psi_t(x;c)\|_{\mathrm{op}} \le L_t$ for all $x \in \mathbb{R}^d$.

Theorem 5 (Location-scale structure of per-step scores)**.** *Under Assumption 4, define* $\mu_t := \Psi_t(\bar{X}_{t-h}; c)$ *and* $\sigma_t := g_t\sqrt{h}\,\|\nabla\Psi_t(\bar{X}_{t-h}; c)\|$. *Conditional on* (X_t, c)*:*

(i) *there exists* $Z_t \sim \mathcal{N}(0,1)$ *(the projection of* ξ *onto the unit gradient direction* $\nabla\Psi_t(\bar{X}_{t-h}; c)/\|\nabla\Psi_t(\bar{X}_{t-h}; c)\|$ *when* $\sigma_t > 0$*; arbitrary when* $\sigma_t = 0$*, in which case the linear term vanishes) such that* $S_t = \mu_t + \sigma_t Z_t + R_t$ *with remainder* $|R_t| \le \frac{1}{2}L_t g_t^2 h\|\xi\|^2$ *a.s. (*Z_t *and* R_t *are not in general independent);*

(ii) *as* $g_t\sqrt{h} \to 0$, $\mathrm{Var}(S_t \mid X_t, c) = \sigma_t^2 + O_{d,L_t,\|\nabla\Psi_t\|}(g_t^3 h^{3/2})$*, where the implicit constant depends on dimension d, the local Hessian bound* L_t*, and the gradient norm at* $\bar{X}_{t-h}$*;*

(iii) *for K i.i.d. noise candidates* $\xi^{(k)} \overset{iid}{\sim} \mathcal{N}(0, I)$ *at step t (random search), let* $S_t^{(k)} := \Psi_t(\bar{X}_{t-h} + g_t\sqrt{h}\,\xi^{(k)}; c)$*. Then*

$$G_t(K) := \mathbb{E}\Big[\max_{1\le k\le K} S_t^{(k)}\,\Big|\,X_t, c\Big] - \mu_t = \sigma_t\, a_K + O_{K,d,L_t}(g_t^2 h), \tag{6}$$

where $a_K := \mathbb{E}[\max(Z_1, \dots, Z_K)]$, $Z_k \overset{iid}{\sim} \mathcal{N}(0,1)$*, with* $a_K \sim \sqrt{2\log K}$*.*

The proof (Appendix B.2) is a second-order Taylor expansion of Ψ_t around $\bar{X}_{t-h}$ (Proposition 9). The product $\sigma_t = g_t\sqrt{h}\,\|\nabla\Psi_t\|$ of diffusion coefficient, step size, and verifier gradient norm is the leading-order coefficient governing how much extra search at t is worth.

Optimal allocation: water-filling. Treating $\{\sigma_t\}$ as a fixed input and dropping the $O(g_t^2 h)$ remainder in Equation (6), summing over t gives the surrogate

$$\max_{\{K_t\}_{t=1}^T} \sum_{t=1}^T \sigma_t\, a_{K_t} \quad \text{s.t.} \sum_{t=1}^T K_t = B,\ K_t \in \mathbb{Z}_{\ge 1}. \tag{7}$$

Write $\Delta a_K := a_{K+1} - a_K$ for the discrete marginal of a_K, and call t *active* at $\{K_t^*\}$ if $K_t^* > 1$.

Theorem 6 (Offline optimality under independence)**.** *Assume* σ_t *depends only on t (not on c or* x_t*) and the remainder* $O_{K,d,L_t}(g_t^2 h)$ *in Theorem 5*(iii) *is uniform in* (c, x_t)*. Then the optimizer* $\{K_t^*\}$ *of Equation* (7) *satisfies:*

(i) Prompt-independence. *The same* $\{K_t^*\}$ *optimizes Equation* (7) *for every* (c, x_t)*.*

(ii) Discrete KKT sandwich. *There exists* $\lambda^* \ge 0$ *(the dual variable for* $\sum_t K_t = B$*) such that, for every active t,*

$$\sigma_t\, \Delta a_{K_t^*-1} \;\ge\; \lambda^* \;\ge\; \sigma_t\, \Delta a_{K_t^*}. \tag{8}$$

(ii') Weak monotonicity.

$$\sigma_{t_1} > \sigma_{t_2} \implies K_{t_1}^* \ge K_{t_2}^*. \tag{9}$$

At the integer level the inequality is strict iff $\sigma_{t_1}/\sigma_{t_2} > \Delta a_{K-1}/\Delta a_K$ *at the binding* K*; the continuous relaxation* $\tilde{a} \in C^1$ *of* a_K *gives strict monotonicity unconditionally.*

(iii) Dominance over uniform. *If* $T \mid B$ *so* $K_t \equiv B/T$ *is feasible,*

$$\sum_{t=1}^T \sigma_t\, a_{K_t^*} \;\ge\; \sum_{t=1}^T \sigma_t\, a_{B/T}, \tag{10}$$

with strict inequality in the continuous relaxation whenever $B > T$ *and* $\sigma_{t_1} \ne \sigma_{t_2}$ *for some pair (at the integer level, conditional on the gap condition in (ii')). Defining the across-timestep dispersion*

$$\mathrm{Var}(\sigma_t) := \frac{1}{T}\sum_{t=1}^T (\sigma_t - \bar{\sigma})^2, \qquad \bar{\sigma} := \frac{1}{T}\sum_t \sigma_t, \tag{11}$$

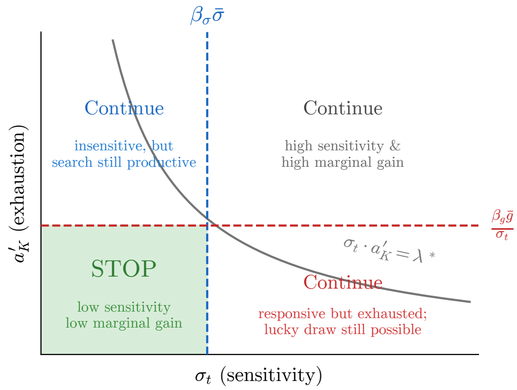

Figure 2: Online stopping decision in the $(\sigma_t, \Delta a_K)$ plane. Horizontal threshold: small recent gain; vertical threshold: small empirical variance (a sensitivity proxy). Stopping triggers only when *both* fire (shaded). The hyperbola $\sigma_t \cdot \Delta a_K = \lambda^*$ is the true optimality boundary; the rectangular proxy is conservative.

the first-order improvement along any budget-neutral pair perturbation $K_{t_1} \to B/T + \epsilon$, $K_{t_2} \to B/T - \epsilon$ *is* $\epsilon(\sigma_{t_1} - \sigma_{t_2})\,\tilde{a}'(B/T) + O(\epsilon^2)$ *(we use* $\mathrm{Var}(\sigma_t)$ *as a summary dispersion measure, without claiming global monotonicity). The case* $B = T$ *forces* $K_t \equiv 1$.

The proof (Appendix B.3) is by Lagrangian first-order conditions and concavity of a_K. The same water-filling structure transfers to any local operator $\mathcal{L}$ satisfying the regularity conditions of Assumption 13, with a_K replaced by an operator-specific gain $\phi_K^{\mathcal{L}}$ (Appendix C).

Regret bound for adaptive scheduling. When $\sigma_t = \sigma_t(c, x_t)$ depends on the prompt, $\{K_t^*\}$ is no longer per-instance optimal. Embed the problem into an abstract sequential-allocation game: at each round t the learner sees the current reward distribution $\mathcal{D}_t$ (in the diffusion problem, the law of $\sigma_t a_{K_t}$), picks $K_t \in \{0, 1\}$ subject to $\sum_t K_t \le B$, and gets reward $\sim \mathcal{D}_t$ if $K_t = 1$ (else 0). Regret is defined against the offline benchmark that knows $\mathcal{D}_1, \ldots, \mathcal{D}_T$ in advance.

Lemma 7 (Regret lower bound for the sequential-allocation game)**.** *For every (possibly randomized) adaptive policy there exists an instance with* $T = 2B$ *such that*

$$\mathbb{E}[R_T] \;\ge\; \tfrac{B}{4} \;=\; \Omega(T). \tag{12}$$

A diffusion policy with prompt-dependent σ_t simulates this game by treating each step as a 0/1 commit, so $\Omega(T)$ regret transfers. The construction (Appendix B.4) sets $\mathcal{D}_t = \delta_{1/2}$ for $t \le B$ and either δ_0 or δ_1 for $t > B$, making the two instances statistically identical through round B. The bound applies to adversarial $\boldsymbol{\sigma}$; Remark 8 below governs the stochastic-$\boldsymbol{\sigma}$ regime targeted by GAINS.

Remark 8 (Online controller as Jensen-gap recovery)*.* For prompt-dependent $\boldsymbol{\sigma} = \boldsymbol{\sigma}(c, x_t)$, define the static oracle value

$$V^*(\boldsymbol{\sigma}) := \max_{\substack{\sum_t K_t = B \\ K_t \ge 1}} \sum_t \sigma_t \, a_{K_t}. \tag{13}$$

V^* is convex in $\boldsymbol{\sigma}$ (pointwise maximum of linear functions), so by Jensen's inequality

$$J(\boldsymbol{\sigma}) \;:=\; \mathbb{E}_{\boldsymbol{\sigma}}[V^*(\boldsymbol{\sigma})] - V^*(\bar{\boldsymbol{\sigma}}) \;\ge\; 0, \tag{14}$$

where $\bar{\boldsymbol{\sigma}} := \mathbb{E}\,\boldsymbol{\sigma}$. We call $J(\boldsymbol{\sigma})$ the *Jensen gap*: the expected gain of an oracle that allocates per prompt over one tuned to the population-mean profile. Lemma 7 prevents closing J in the adversarial worst case, but structured $\boldsymbol{\sigma}$ admits partial recovery. GAINS stops at step t only when both the recent gain $\widetilde{g}^{(j)}$ and the within-step variance $\sigma^{(j)}$ are small (verifier locally insensitive *and* search no longer improving); either signal alone keeps search active. Figure 2 illustrates the rule. The convexity argument carries over to general operators (Appendix C).

Table 1: SD results across NFE budgets.

NFE	Uniform	GAINS
Brightness		
400	0.7025 ± 0.047	$\mathbf{0.7248\pm0.056}$
500	0.7094 ± 0.063	$\mathbf{0.7346\pm0.060}$
800	0.7511 ± 0.062	$\mathbf{0.7832\pm0.046}$
Compressibility		
400	0.8833 ± 0.017	$\mathbf{0.8946\pm0.016}$
500	0.8825 ± 0.017	$\mathbf{0.9004\pm0.015}$
800	0.8892 ± 0.020	$\mathbf{0.9008\pm0.012}$

Table 2: EDM results across NFE budgets.

NFE	Uniform	GAINS
Brightness		
144	0.9507 ± 0.015	$\mathbf{0.9887\pm0.005}$
180	0.9617 ± 0.016	$\mathbf{0.9904\pm0.004}$
288	0.9787 ± 0.018	$\mathbf{0.9988\pm0.001}$
Compressibility		
144	0.6714 ± 0.017	$\mathbf{0.6845\pm0.007}$
180	0.6774 ± 0.012	$\mathbf{0.7003\pm0.009}$
288	0.6901 ± 0.012	$\mathbf{0.7165\pm0.009}$

5 Experiments

We test one prediction from our theory: under fixed NFE, non-uniform allocation should beat uniform whenever per-step sensitivity varies, with gains growing in the dispersion of $\{\sigma_t\}$.

Setting. We evaluate GAINS on Stable Diffusion (SD; Rombach et al., 2022), EDM [Karras et al., 2022], and a flow-based model under strict NFE budgets, with two lightweight verifiers: **Brightness** (luminance) and **Compressibility** (JPEG size). The Uniform benchmark runs the same ϵ-greedy noise search [Ramesh and Mardani, 2025] but allocates evenly. GAINS replaces uniform with offline profiling + online early stopping, redistributing saved NFE so the realized total exactly matches B. All runs use a single A100.

Hyperparameters and reproducibility. EDM ($T = 18$): $\epsilon = 0.4$, $\lambda = 0.15$, $N = 4$, $\beta_g = 0.3$, $\beta_\sigma = 0.7$, $\delta = 2$. SD ($T = 50$): $N = 4$, $\beta_g = 0.1$, $\beta_\sigma = 0.8$. All runs use the *revert-on-negative* rule with $W_g = 4$; $\{K_t\}$ from offline profiling. Each configuration averages 20 seeds; SD prompt-set runs use 20 prompts $\times$ 10 repeats. Code released on acceptance.

5.1 Stable Diffusion and EDM: Scaling with NFE

GAINS improves both absolute scores and the *rate* at which quality scales with NFE (Tables 1 and 2). On SD, the Brightness margin widens from $+0.0223$ (400 NFE) to $+0.0321$ (800 NFE); GAINS/400 beats Uniform/500 (**20%** fewer) and matches Uniform/800 on Compressibility (**50%** fewer). On EDM, GAINS/144 matches or beats Uniform/288 (**50%/37.5%** fewer); the larger EDM margin tracks Theorem 6(iii), since EDM's $\{\sigma_t\}$ concentrate on a middle segment with higher $\mathrm{Var}(\sigma_t)$ than SD's early-spread profile. Seed-to-seed std also tightens (±0.005 vs. ±0.015 at EDM/144).

5.2 Ablation, Generalization, Operator and Flow Compatibility

Four stress tests (full tables in Appendix D) confirm the gains. **(a) Online vs. offline-only** (SD/400 NFE): online adds $+0.0090$/$+0.0073$ on Brightness/Compressibility (Jensen-gap mechanism, Remark 8). **(b) Generalization** (20 prompts $\times$ 10 repeats): Brightness $0.6949 \to 0.7213$, Compressibility $0.7938 \to 0.8076$. **(c) Operator compatibility**: gains hold under both zero-order and random search (Remark 16). **(d) Flow-based models**: after stochasticity injection [Singh and Fischer, 2024, Kim et al., 2025], GAINS adds $+0.005$ to $+0.015$ across $\{80, 100, 150\}$ NFE, smaller than $+0.02$ to $+0.04$ on SD/EDM (narrower $\{\sigma_t\}$ dispersion after ODE-to-SDE; Appendix D.4).

6 Conclusion

We shifted inference-time compute allocation from *how* to search at one step to *where* along the trajectory. The two-level framework (Section 3) recovers plain sampling, best-of-N, ϵ-greedy, and particle methods as special cases; GAINS pairs offline profiling with online feedback, with water-filling (Theorem 6) and an $\Omega(T)$ regret bound (Lemma 7) justifying the hybrid; it matches uniform with 20–50% fewer NFE on SD/EDM/flow. *Limitations:* leading-order in $g_t\sqrt{h}$, $\Psi_t \in C^2$; the MDP formulation (Appendix A) opens learned schedulers and multi-verifier composition.

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A General MDP Formulation of Noise Trajectory Search

This appendix has two purposes. First, it presents a general Markov decision process (MDP) for inference-time search over diffusion-noise trajectories, extending the simpler formulation used in the main text. Second, it uses this MDP as a unified language for classifying ten prior inference-time-search methods (Table 3) and locating where GAINS sits among them.

The main-text framework (Section 3) uses a restricted policy: the sampler advances through timesteps in a single forward pass, and each timestep runs the same local search operator (e.g., random search) over its noise draw. The general MDP introduced here lifts both restrictions, allowing (i) multiple verifier types of different cost, (ii) backtracking to revisit earlier timesteps, and (iii) different search operators at different timesteps. Ramesh and Mardani [2025] first cast diffusion sampling as an MDP with state (c, x_t, t) and terminal reward $r(x_0, c)$, then relaxed it to independent contextual bandits for tractability. We extend this formulation in four respects: the state includes the search history $\mathcal{H}_t$ and the remaining compute budget R; the action set is partitioned into Search, Verify, and Move categories (defined below); the reward admits weighted combinations of multiple verifier scores; and we use the resulting MDP to classify ten existing methods as special-case policies.

State space. At each decision step, the agent observes $s = (c, t, x_t, \mathcal{H}_t, R)$. Here c is the prompt or other conditioning signal, $t \in \{1, \ldots, T\}$ indexes the current diffusion timestep (counting down from T to 0), x_t is the current latent at that timestep, $\mathcal{H}_t$ records the search history at t (past candidate noises, verifier scores, and actions taken), and $R \in \{0, \ldots, B\}$ is the remaining compute budget measured in number of function evaluations (NFE) of the diffusion network. The initial state is $s_0 = (c, T, x_T, \emptyset, B)$ with $x_T \sim \mathcal{N}(0, I)$, and the episode terminates when either $t = 0$ (a clean sample is reached) or $R = 0$ (the budget is exhausted).

Action space. The action set partitions into three categories, $\mathcal{A}(s) = \mathcal{A}_{\text{search}} \cup \mathcal{A}_{\text{verify}} \cup \mathcal{A}_{\text{move}}$, distinguished by whether they propose a noise candidate, score it, or commit to a transition:

- *Search* ($\mathcal{A}_{\text{search}}$): one iteration of a local search operator over the noise variable ε_t at the current timestep t, without changing t or x_t. Variants include RANDOMSEARCH (draw $\varepsilon_t \sim \mathcal{N}(0, I)$, keep if it scores higher than the incumbent), EPSGREEDY (with probability ϵ explore via a fresh Gaussian draw, otherwise exploit by perturbing the incumbent), ZERO-ORDER (a zero-order finite-difference gradient step on the noise variable), and LANGEVIN (one Langevin MCMC step over the noise variable). Cost: 1 NFE per iteration.
- *Verify* ($\mathcal{A}_{\text{verify}}$): evaluate the current x_t or its one-step prediction of the clean sample $\hat{x}_0$, without modifying the latent. VERIFYLIGHT reuses the already-computed $\hat{x}_0$ and applies a cheap verifier (e.g., JPEG compressibility, pixel-level statistics, or the LAION aesthetic predictor) at 0 additional NFE. VERIFYHEAVY runs additional passes of the diffusion denoising network to refine the $\hat{x}_0$ estimate before applying the verifier, at ≥ 1 NFE.
- *Move* ($\mathcal{A}_{\text{move}}$): commit to a timestep transition. FORWARD sets $x_{t-1} = F_\theta(x_t, t, c, \varepsilon_t^*)$ using the best noise candidate ε_t^* found so far at t and advances the index from t to $t-1$. We charge FORWARD at 0 NFE: the forward pass through the denoising network has already been performed when ε_t^* was scored as a SEARCH candidate, and FORWARD merely commits the cached latent. BACKTRACK returns from t to $t + 1$, restores the previously saved latent, retains $\mathcal{H}_t$, and also costs 0 NFE. With this convention the per-step NFE accounting matches the main text: a step at which K_t SEARCH actions are taken (followed by one FORWARD) consumes exactly K_t NFE.

Transition dynamics and reward. The transitions follow directly from the action descriptions: after a SEARCH action, t and x_t are unchanged, $\mathcal{H}_t$ is augmented with the new candidate, and R

Table 3: Existing inference-time search methods as policies in the noise-trajectory-search MDP. $\mathcal{A}_{\text{used}}$ lists the action types each method invokes (FWD = forward move, RAND = random search, EPSG = ϵ-greedy search, ZERO = zero-order search, LANG = Langevin search, ANY = any operator satisfying Assumption 13). **Backtrack?** indicates whether the policy ever returns to an earlier timestep. **Schedule** describes how the per-timestep search budget is allocated.

Method	$\mathcal{A}_{\text{used}}$	Backtrack?	Schedule
Plain sampling	FWD	No	$K_t = 1\,\forall t$
Best-of-N [Ma et al., 2025]	FWD	No	N full trajectories
ϵ-Greedy [Ramesh and Mardani, 2025]	EPSG, FWD	No	Uniform $K_t = K$
DNO [Tang et al., 2024]	ZERO, FWD	No	Uniform $K_t = K$
RBF [Kim et al., 2025]	RAND, FWD	No	Online rollover
T-SCEND [Zhang et al., 2025b]	RAND, FWD	No	Fixed 2-phase
ABCD [Lee et al., 2025]	RAND, FWD	Yes	Online adaptive
Tree/BFS [Zhang et al., 2025c]	LANG, FWD	Yes	Tree expansion
FK Steering [Singhal et al., 2025]	RAND, FWD	No	Non-uniform (SMC)
VT [Sundaresha et al., 2025]	RAND, FWD	No	Online threshold
GAINS (this work)	ANY, FWD	No	Offline + online

decreases by 1; after FORWARD, t decreases by 1, x_{t-1} is set to $F_\theta(x_t, t, c, \varepsilon_t^*)$, and R is unchanged; after BACKTRACK, t increases by 1, x_{t+1} is restored from saved history, and $\mathcal{H}$ is retained. The reward is non-zero only at termination: $r(s_{\text{terminal}}) = v(x_0, c)$, where v is a verifier score on the final clean sample (weighted combinations $v = \alpha v_1 + \beta v_2$ of multiple verifiers are also allowed). The agent maximizes expected terminal score, $\max_\pi \mathbb{E}[v(x_0, c)]$, subject to the budget constraint $R \geq 0$ throughout.

Existing methods as MDP policies. Table 3 classifies prior inference-time-search methods as policies in the MDP defined above. Three structural patterns emerge. (i) *No method mixes verifier granularity:* all methods commit a priori to either a cheap or an expensive verifier; adaptively switching between VERIFYLIGHT and VERIFYHEAVY based on state remains unexplored. (ii) *Backtracking is rare:* only ABCD and tree-search methods ever return to an earlier timestep $t+1$ from t. (iii) *Operator selection is static:* each listed method fixes one search operator (random, ϵ-greedy, zero-order, or Langevin) for the entire trajectory rather than choosing per timestep.

Main-text formulation as a special case. The two-level framework of Section 3 and the GAINS algorithm (Algorithm 1) correspond to a restricted policy class within this MDP. The restrictions are: no backtracking, a single search operator throughout the trajectory, a single verifier, and a factored budget rule in which the total budget B is pre-partitioned across timesteps by offline profiling and then adjusted online within a small slack δ. In MDP terms, the policy's dependence on the history $\mathcal{H}_t$ collapses to two running summaries at the current timestep: the running mean gain $\overline{g}^{\,\text{run}}$ (improvement of the verifier score so far) and the running mean of within-step empirical variance $\overline{\sigma}^{\,\text{run}}$ (dispersion of candidate scores so far; distinct from the timestep-average $\bar{\sigma}$ of theoretical sensitivities in Theorem 6). Relaxing these restrictions yields richer policy classes inside the same MDP that may further improve the quality-compute frontier, but learning such policies likely requires reinforcement learning rather than the closed-form scheduling rules of the main text.

B Proofs

This appendix collects complete proofs of the theoretical results stated in Section 4.3. The proofs are presented in the same order as the main text. We begin with an auxiliary Taylor expansion of the verifier score, then prove the location-scale structure (Theorem 5), the offline water-filling characterization (Theorem 6), and finally the worst-case regret lower bound (Lemma 7).

B.1 Auxiliary: Taylor Expansion of the Verifier Score

Before turning to Theorem 5, we record a one-step Taylor expansion of the verifier score around the deterministic Euler point. This expansion is the source of the linear-in-noise leading term that drives the location-scale structure.

Proposition 9 (Score Taylor expansion)**.** *Under Assumption 4, conditional on (X_t, c),*

$$S_t \;=\; \Psi_t(\bar{X}_{t-h};\, c) \;+\; g_t\sqrt{h}\,\big\langle \nabla\Psi_t(\bar{X}_{t-h};\, c),\, \xi \big\rangle \;+\; R_t, \tag{15}$$

where the remainder satisfies $|R_t| \le \frac{1}{2}\, L_t\, g_t^2\, h\, \|\xi\|^2$ almost surely, and hence $\mathbb{E}[\,|R_t| \mid X_t, c\,] \le \frac{1}{2}\, L_t\, g_t^2\, h\, d$.

Proof. The Euler-Maruyama update gives $X_{t-h} = \bar{X}_{t-h} + g_t\sqrt{h}\,\xi$, so $\Psi_t(X_{t-h}; c)$ is the verifier score evaluated at a Gaussian perturbation of the deterministic point $\bar{X}_{t-h}$. A second-order Taylor expansion of $\Psi_t(\cdot\,; c)$ around $\bar{X}_{t-h}$ gives

$$\Psi_t(X_{t-h}; c) = \Psi_t(\bar{X}_{t-h}; c) + \big\langle \nabla\Psi_t(\bar{X}_{t-h}; c),\, X_{t-h} - \bar{X}_{t-h} \big\rangle + R_t,$$

where $R_t = \frac{1}{2}(X_{t-h} - \bar{X}_{t-h})^\top \nabla^2\Psi_t(\widetilde{X}; c)\,(X_{t-h} - \bar{X}_{t-h})$ for some $\widetilde{X}$ on the segment joining $\bar{X}_{t-h}$ and X_{t-h}. Substituting $X_{t-h} - \bar{X}_{t-h} = g_t\sqrt{h}\,\xi$ yields Equation (15). The remainder bound follows from

$$|R_t| \le \tfrac{1}{2}\|\nabla^2\Psi_t(\widetilde{X}; c)\|_{\mathrm{op}}\, \|g_t\sqrt{h}\,\xi\|^2 \le \tfrac{1}{2} L_t\, g_t^2\, h\, \|\xi\|^2,$$

using Assumption 4. Taking conditional expectations and using $\mathbb{E}[\|\xi\|^2 \mid X_t, c] = d$ completes the proof. $\square$

B.2 Proof of Theorem 5

The proof has three parts. Part (i) reduces the verifier score at a noisy candidate to a one-dimensional Gaussian by projecting the noise onto the gradient direction. Part (ii) controls the conditional variance using the remainder bound from Proposition 9. Part (iii) lifts the single-candidate result to the maximum over K candidates and identifies the dependence on K through the Gaussian order-statistic constant a_K.

Part (i): location-scale form. When $\|\nabla\Psi_t(\bar{X}_{t-h}; c)\| > 0$, write

$$\langle \nabla\Psi_t, \xi \rangle = \|\nabla\Psi_t\|\, Z_t, \qquad Z_t := \langle \nabla\Psi_t / \|\nabla\Psi_t\|,\, \xi \rangle.$$

Conditional on (X_t, c), the unit vector $\nabla\Psi_t/\|\nabla\Psi_t\|$ is deterministic, and $\xi \sim \mathcal{N}(0, I)$. The projection of an isotropic Gaussian onto a fixed unit vector is standard normal, so $Z_t \sim \mathcal{N}(0, 1)$. Substituting into Proposition 9 yields $S_t = \mu_t + \sigma_t Z_t + R_t$ with $\sigma_t = g_t\sqrt{h}\,\|\nabla\Psi_t\|$. In the degenerate case $\nabla\Psi_t(\bar{X}_{t-h}; c) = 0$, we have $\sigma_t = 0$ and the linear term vanishes, so the result holds trivially with Z_t defined arbitrarily.

Part (ii): variance approximation. We now show that the conditional variance of S_t is dominated by the linear term $\sigma_t Z_t$. Decompose

$$\mathrm{Var}(S_t \mid X_t, c) = \mathrm{Var}(\sigma_t Z_t + R_t) = \sigma_t^2 + 2\ \mathrm{Cov}(\sigma_t Z_t, R_t) + \mathrm{Var}(R_t).$$

By Cauchy-Schwarz,

$$|\operatorname{Cov}(\sigma_t Z_t, R_t)| \le \sqrt{\mathrm{Var}(\sigma_t Z_t)}\,\sqrt{\mathrm{Var}(R_t)} \le \sigma_t\,\sqrt{\mathbb{E}[R_t^2]}.$$

Using Proposition 9, $|R_t| \le \frac{1}{2} L_t g_t^2 h \|\xi\|^2$, hence

$$\mathbb{E}[R_t^2 \mid X_t, c] \le \tfrac{1}{4} L_t^2 g_t^4 h^2\, \mathbb{E}[\|\xi\|^4] = \tfrac{1}{4} L_t^2 g_t^4 h^2\, d(d+2),$$

because $\xi \sim \mathcal{N}(0, I)$ implies $\mathbb{E}[\|\xi\|^4] = d(d+2)$. Therefore

$$|\operatorname{Cov}(\sigma_t Z_t, R_t)| \le \tfrac{1}{2} L_t \sqrt{d(d+2)}\, \sigma_t g_t^2 h = O_{d, L_t, \|\nabla\Psi_t\|}(g_t^3 h^{3/2}),$$

where the implicit constant absorbs L_t and $\|\nabla\Psi_t\|$. Likewise,

$$\mathrm{Var}(R_t) \le \mathbb{E}[R_t^2] \le \tfrac{1}{4} L_t^2 g_t^4 h^2\, d(d+2) = O_{d, L_t}(g_t^4 h^2),$$

which is lower order than the covariance term as $g_t\sqrt{h} \to 0$. Combining yields $\mathrm{Var}(S_t \mid X_t, c) = \sigma_t^2 + O_{d, L_t, \|\nabla\Psi_t\|}(g_t^3 h^{3/2})$.

Part (iii): expected gain factorization. Apply Part (i) to each of the K independent candidates: $S_t^{(k)} = \mu_t + \sigma_t Z_t^{(k)} + R_t^{(k)}$, where $Z_t^{(k)} = \langle \nabla\Psi_t/\|\nabla\Psi_t\|, \xi^{(k)}\rangle \sim \mathcal{N}(0,1)$ are i.i.d. across k. Taking the maximum,

$$\max_k S_t^{(k)} = \mu_t + \max_k\big(\sigma_t Z_t^{(k)} + R_t^{(k)}\big).$$

Using the elementary inequality $\big|\max_k(a_k+b_k) - \max_k a_k\big| \le \max_k |b_k|$, we obtain

$$\big|\max_k S_t^{(k)} - (\mu_t + \sigma_t \max_k Z_t^{(k)})\big| \le \max_k |R_t^{(k)}|.$$

Since $\max_k |R_t^{(k)}| \le \sum_k |R_t^{(k)}|$, taking conditional expectations gives

$$\mathbb{E}\big[\max_k |R_t^{(k)}| \mid X_t, c\big] \le K \cdot \tfrac{1}{2} L_t g_t^2 h\, d.$$

Therefore

$$G_t(K) = \mathbb{E}\big[\max_k S_t^{(k)} \mid X_t, c\big] - \mu_t = \sigma_t\, a_K + O_{K,d,L_t}(g_t^2 h),$$

where the implicit constant in $O_{K,d,L_t}(\cdot)$ depends on K, the dimension d, and the Hessian bound L_t from Assumption 4, but not on the diffusion coefficient g_t or the verifier gradient $\nabla\Psi_t$. Substituting $\sigma_t = g_t\sqrt{h}\,\|\nabla\Psi_t(\bar{X}_{t-h};c)\|$ yields the explicit form. □

Remark 10 (Sensitivity-scaling mechanism). Theorem 5 identifies the per-step sensitivity σ_t as the product of three quantities: the diffusion coefficient g_t, the discretization step $\sqrt{h}$, and the verifier gradient norm $\|\nabla\Psi_t(\bar{X}_{t-h};c)\|$ at the deterministic Euler point. Timesteps with large g_t (strong noise injection) or large $\|\nabla\Psi_t\|$ (large verifier gradient) therefore benefit most from noise search, which explains the non-uniform sensitivity patterns observed in offline profiling. The C^2 smoothness assumption on Ψ_t amounts to requiring that both the denoising network D_θ and the verifier v are twice differentiable, which holds for neural networks with standard smooth activation functions. The per-timestep Hessian bound L_t is typically larger at early timesteps, where the verifier score function has more curvature, and smaller at late timesteps, where denoised predictions are nearly converged.

Corollary 11 (SDE-based approximate allocation)**.** *Under Assumption 4, with step size h and diffusion coefficients $\{g_t\}_{t=1}^T$, for any fixed budget allocation $\{K_t\}_{t=1}^T$, the following equality holds conditional on the trajectory $\{X_t\}_{t=1}^T$ (equivalently, almost surely along each realized path):*

$$\sum_{t=1}^T G_t(K_t) = \sum_{t=1}^T g_t\sqrt{h}\,\|\nabla\Psi_t(\bar{X}_{t-h};c)\|\, a_{K_t} + \sum_{t=1}^T O_{K_t,d,L_t}(g_t^2 h). \tag{16}$$

Dropping the second-order remainder reduces the offline allocation problem to a concave resource allocation with per-step weights $\sigma_t = g_t\sqrt{h}\,\|\nabla\Psi_t\|$, which is the formulation (7) *in Section 4.3.*

B.3 Proof of Theorem 6

The proof has three parts. Part (i) shows that the prompt-independence assumption collapses the per-prompt allocation into a single deterministic problem. Part (ii) derives a discrete KKT sandwich via an exchange argument and establishes weak monotonicity $\sigma_{t_1} > \sigma_{t_2} \Rightarrow K^*_{t_1} \ge K^*_{t_2}$ at the integer level. Part (iii) shows that the uniform allocation $K_t \equiv B/T$ is strictly suboptimal whenever the sensitivities $\{\sigma_t\}$ are not all equal.

Part (i): prompt-independence. Under the theorem's assumption that σ_t depends only on the timestep index, the leading-order term $\sigma_t a_{K_t}$ in $G_t(K_t)$ depends only on (t, K_t). By the additional uniformity assumption that the $O_{K,d,L_t}(g_t^2 h)$ remainder in Theorem 5(iii) is bounded uniformly in (c, x_t), the surrogate objective $\sum_t \sigma_t a_{K_t}$ in Equation (7) is the same function of $\{K_t\}$ across all prompts and initial states, up to a uniform $O(g_t^2 h \cdot T)$ remainder absorbed in the dropping. Any $\{K_t^*\}$ solving Equation (7) is therefore simultaneously optimal for the surrogate across all prompts.

Part (ii): discrete KKT sandwich and weak monotonicity. We work directly with the integer problem via a discrete-exchange argument. By concavity of a_K (David and Nagaraja 2003, Ch. 4) restricted to $K \ge 1$, the marginal gain $\Delta a_K := a_{K+1} - a_K$ is non-increasing for $K \ge 1$. The location-scale structure makes a_K the same function across all timesteps, so Δa_K is t-independent.

Discrete KKT sandwich. At any optimum $\{K_t^*\}$, define the discrete marginal price $\lambda^* := \min_{t:\, K_t^*>1} \sigma_t\, \Delta a_{K_t^*-1}$ (or $\lambda^* := 0$ if all $K_t^* = 1$). For every active timestep ($K_t^* > 1$), the inequality

$$\sigma_t\, \Delta a_{K_t^*-1} \;\geq\; \lambda^* \;\geq\; \sigma_t\, \Delta a_{K_t^*}$$

holds: the lower bound is by definition of λ^*. For the upper bound, let $s \in \arg\min_{r:\, K_r^*>1} \sigma_r\, \Delta a_{K_r^*-1}$ achieve λ^*, and consider the *reverse* exchange: move one unit from s to t (feasible since $K_s^* > 1$, and budget-neutral). The objective change is $\sigma_t \Delta a_{K_t^*} - \sigma_s \Delta a_{K_s^*-1}$, which must be ≤ 0 at optimality, yielding $\sigma_t \Delta a_{K_t^*} \leq \sigma_s \Delta a_{K_s^*-1} = \lambda^*$. For inactive timesteps ($K_t^* = 1$) only the right inequality $\sigma_t\, \Delta a_1 \leq \lambda^*$ is required (otherwise moving a unit *into* t would improve).

Weak monotonicity. Suppose $\sigma_{t_1} > \sigma_{t_2}$ but $K_{t_1}^* < K_{t_2}^*$ for some pair. The swap $(K_{t_1}^*, K_{t_2}^*) \to (K_{t_1}^*+1, K_{t_2}^*-1)$ (feasible since $K_{t_2}^* \geq 2$) changes the objective by

$$\sigma_{t_1}\, \Delta a_{K_{t_1}^*} - \sigma_{t_2}\, \Delta a_{K_{t_2}^*-1} > 0,$$

since $\Delta a_{K_{t_1}^*} \geq \Delta a_{K_{t_2}^*-1}$ (because $K_{t_1}^* + 1 \leq K_{t_2}^*$ and Δa is non-increasing for $K \geq 1$) and $\sigma_{t_1} > \sigma_{t_2}$. This contradicts optimality, so $K_{t_1}^* \geq K_{t_2}^*$.

Strictness at the integer level. The integer problem does *not* guarantee strict monotonicity even when both timesteps are interior. Concretely, if $\sigma_{t_1}/\sigma_{t_2} \leq \Delta a_{K-1}/\Delta a_K$ at the relevant K, the sandwich admits $K_{t_1}^* = K_{t_2}^* = K$. (Example: $T = 2$, $B = 6$, $\sigma_1 = 1.01$, $\sigma_2 = 1$; the optimum is $(3, 3)$.) The continuous relaxation $a_K \mapsto \tilde{a}_K \in C^1([1, \infty))$ recovers strict monotonicity unconditionally, but the integer optimum can have ties.

Steps stuck at the floor $K_t^* = 1$ may have arbitrary σ_t ordering: a small budget B relative to T can pin many timesteps at the floor regardless of their sensitivity. (Example: $T = 3$, $B = 4$, $(\sigma_1, \sigma_2, \sigma_3) = (10, 1, 0.5)$ optimally allocates $(2, 1, 1)$, even though $\sigma_2 > \sigma_3$.) Weak monotonicity concerns only the active set.

Part (iii): dominance of non-uniform allocation. We separate two cases.

Case $B = T$. The constraints $\sum_t K_t = B$ and $K_t \geq 1$ force $K_t \equiv 1$, so the uniform allocation is the unique feasible point and dominates trivially. There is no room for reallocation, regardless of $\{\sigma_t\}$.

Case $B > T$ (continuous relaxation). Uniform allocation $K_t \equiv B/T$ is feasible only when $T \mid B$; otherwise the closest integer uniform is feasible up to a ± 1 rounding. The strictness statement in the theorem is a property of the continuous relaxation, in which $\{K_t\}$ are allowed to range over $[1, \infty)^T$. To work in this relaxation, extend a_K to a C^1 strictly concave function $\tilde{a}$ on $[1, \infty)$ that interpolates the integer values $\{a_K\}_{K \geq 1}$. Such an extension exists because the discrete first-differences $\{\Delta a_K\}_{K \geq 1}$ form a strictly decreasing positive sequence (David and Nagaraja 2003, Ch. 4): any smooth interpolant whose derivative is strictly decreasing and matches Δa_K in average across each unit interval works (e.g., a C^1 Hermite spline with derivative values $\tilde{a}'(K) := \frac{1}{2}(\Delta a_{K-1} + \Delta a_K)$ at each integer $K \geq 2$ and $\tilde{a}'(1) := \Delta a_1$). At the uniform point, the marginal gains are $\tilde{G}_t'(B/T) = \sigma_t\, \tilde{a}'(B/T)$. Whenever $\sigma_{t_1} \neq \sigma_{t_2}$ for some pair (t_1, t_2), these marginals differ, so the stationarity condition from Part (ii) is violated. Because $B > T$ (with both integers), $B/T \geq 1 + 1/T > 1$, so the lower-bound constraint $K_t \geq 1$ is inactive at the uniform point, and the budget-neutral perturbation $K_{t_1} = B/T + \epsilon$, $K_{t_2} = B/T - \epsilon$ stays feasible for small $\epsilon > 0$. Expanding to first order,

$$\begin{aligned} &\big[\tilde{G}_{t_1}(B/T + \epsilon) + \tilde{G}_{t_2}(B/T - \epsilon)\big] - \big[\tilde{G}_{t_1}(B/T) + \tilde{G}_{t_2}(B/T)\big] \\ &\quad = \epsilon\, (\sigma_{t_1} - \sigma_{t_2})\, \tilde{a}'(B/T) + O(\epsilon^2) > 0 \end{aligned}$$

when $\sigma_{t_1} > \sigma_{t_2}$ (relabel if needed), where the common factor $\tilde{a}'(B/T)$ can be extracted because the location-scale structure makes a_K (and hence $\tilde{a}$) independent of t. Hence the uniform allocation is not a stationary point of the continuous relaxation, and strict concavity of $\tilde{a}$ together with concavity of the constraint set yields strict suboptimality: $\sum_t \sigma_t\, \tilde{a}(K_t^\star) > \sum_t \sigma_t\, \tilde{a}(B/T)$ for the continuous optimizer $\{K_t^\star\}$ when $\{\sigma_t\}$ are not all equal. The local first-order improvement at the uniform point scales linearly in any pairwise σ-gap $|\sigma_{t_1} - \sigma_{t_2}|$ (a directional, not global, statement); we adopt the across-timestep variance as our dispersion measure but do not claim a global monotonicity of the gap in this measure. The integer optimum can fall short of the continuous one by a discrete-marginal step (cf. the example in Part (ii) where ties at the integer level absorb small σ-gaps); strictness in the integer problem is therefore conditional on the σ-gap exceeding a discrete-marginal step, while the continuous relaxation gives strictness unconditionally. □

B.4 Proof of Lemma 7

The proof proceeds by an indistinguishability argument. We construct two problem instances that are statistically identical over the first B rounds, so that any adaptive policy must commit to the same expected first-half spending on both. We then show that the two continuations have opposite optimal allocations, forcing linear regret on at least one instance.

Fix any adaptive policy and let $T = 2B$. For $t = 1, \ldots, B$, both instances set $\mathcal{D}_t = \delta_{1/2}$, the Dirac mass at reward $1/2$. Each unit allocated in this first half therefore returns deterministic reward $1/2$.

The two instances diverge in the second half $t = B+1, \ldots, 2B$:

$$\text{Instance A: } \mathcal{D}_t = \delta_0, \qquad \text{Instance B: } \mathcal{D}_t = \delta_1.$$

Let $S = \mathbb{E}\big[\sum_{t=1}^{B} K_t\big]$ denote the expected budget the policy spends in the first half. The learner's only observations during rounds $1, \ldots, B$ are the distributions $\mathcal{D}_t$, which agree across the two instances, so its expected first-half allocation S is the same on both instances.

Instance A. The second half returns zero reward, so the offline benchmark spends all B units in the first half, achieving total reward $B/2$. The learner's expected reward is $S/2$, since each first-half unit returns $1/2$ and second-half units return 0. The expected regret is therefore

$$\mathbb{E}[R_T] \;\geq\; \frac{B}{2} - \frac{S}{2}.$$

Instance B. The second half returns reward 1 per unit, so the offline benchmark spends all B units there, achieving total reward B. The learner spends S in expectation in the first half (expected reward $S/2$) and at most $B - S$ in expectation in the second half (expected reward at most $B - S$), giving total expected reward at most $S/2 + (B - S) = B - S/2$. The expected regret is therefore

$$\mathbb{E}[R_T] \;\geq\; B - \Big(B - \frac{S}{2}\Big) = \frac{S}{2}.$$

Combining the two cases, for any adaptive policy with expected first-half spending $S \in [0, B]$,

$$\mathbb{E}[R_T] \;\geq\; \max\Big\{\frac{B-S}{2},\, \frac{S}{2}\Big\} \;\geq\; \frac{B}{4} = \Omega(T),$$

where the last inequality holds because $\max\{(B-S)/2, S/2\} \geq B/4$ for all $S \in [0, B]$, with the minimum attained at $S = B/2$. Since the bound applies for every choice of S, at least one of the two instances satisfies $\mathbb{E}[R_T] \geq B/4$, which establishes the worst-case lower bound. □

Remark 12 (Interpretation of the lower bound). Lemma 7 establishes that linear regret is unavoidable for any fully adaptive policy when the underlying distributions $\{\mathcal{D}_t\}$ are drawn from a worst-case adversarial family. The construction exploits hidden information in the second half of the trajectory. Until round t is reached, the learner cannot distinguish Instance A from Instance B based on the observable history, so any commitment to spend in the first half (which is correct when the future is bad) or to save for the second half (which is correct when the future is good) is provably suboptimal on the other instance. In the diffusion setting, this means a worst-case adversarial choice of per-prompt sensitivities $\boldsymbol{\sigma}(c, x_t)$ would force linear regret on any online controller. The Jensen-gap argument (Remark 8) sidesteps this worst case by exploiting the convexity of V^* in $\boldsymbol{\sigma}$: when sensitivities are drawn from a structured (non-adversarial) distribution, the online controller recovers a positive fraction of the gap, as confirmed empirically by the offline-vs-online ablation in Table 5.

C Extension to General Local Search Operators

This appendix extends the scheduling theory of Section 4.3 from random search to a broader family of local search operators. The main-text analysis instantiates the local operator as pure random search (i.i.d. Gaussian noise draws, keep-best). The scheduling results, however, only use three properties of the resulting gain sequence: it is increasing in the per-step budget K, its marginal increments do not accelerate, and its functional form does not depend on the diffusion timestep t. Any local search operator that produces a gain sequence with these three properties inherits the same offline water-filling allocation and the same Jensen-gap argument for online adaptation. Below we make this observation precise. We first state four regularity conditions on a generic local operator (Assumption 13), prove a general gain factorization (Theorem 15), transfer the offline and online scheduling results (Corollaries 17 and 19), and then verify the conditions for ϵ-greedy exploration and local perturbation search alongside random search.

C.1 Abstract Local Search Framework

This subsection sets up a generic local search operator and the gain it produces, then states the four regularity conditions (A1)–(A4) that the rest of the appendix needs.

A local search operator $\mathcal{L}$ runs at a fixed diffusion timestep t and produces K candidate noise iterates $\xi^{(1)}, \ldots, \xi^{(K)}$ in $\mathbb{R}^d$, each evaluated through the diffusion transition. Random search draws each $\xi^{(j)} \sim \mathcal{N}(0, I)$ independently; iterative operators (ϵ-greedy, local perturbation) may make $\xi^{(j+1)}$ depend on $\xi^{(1)}, \ldots, \xi^{(j)}$. The *gain* from K iterations measures how much the best-of-K verifier score exceeds the no-search baseline μ_t:

$$G_t^{\mathcal{L}}(K) := \mathbb{E}\Big[\max_{1 \le j \le K} \Psi_t\big(\bar{X}_{t-h} + g_t\sqrt{h}\,\xi^{(j)};\, c\big) \,\Big|\, X_t, c\Big] - \mu_t, \tag{17}$$

where $\mu_t = \Psi_t(\bar{X}_{t-h}; c)$ is the score with no search (as in the main text), and we adopt the convention $G_t^{\mathcal{L}}(0) := 0$. This convention aligns with the main-text accounting: a step that consumes K_t NFE evaluates exactly K_t noise candidates. The next assumption collects four properties that suffice for every result in this appendix.

Assumption 13 (Local search regularity)**.** Define the projected gain sequence

$$\phi_K^{\mathcal{L}} := \mathbb{E}\big[\max_{1 \le j \le K} \langle u, \xi^{(j)} \rangle\big] \tag{18}$$

for any unit direction u (the value is shown to be u-independent below); set $\phi_0^{\mathcal{L}} := 0$. A local search operator $\mathcal{L}$ satisfies:

(A1) *Strict monotone improvement.* $\phi_K^{\mathcal{L}}$ is strictly increasing in K for $K \geq 1$.

(A2) *Diminishing returns.* The marginal gains $\Delta\phi_K^{\mathcal{L}} := \phi_{K+1}^{\mathcal{L}} - \phi_K^{\mathcal{L}}$ are non-increasing for $K \geq 1$ (equivalently, $\phi_K^{\mathcal{L}}$ is discretely concave on $K \geq 1$). (We exclude $K = 0$ because $\phi_1^{\mathcal{L}} = \phi_0^{\mathcal{L}} = 0$ for operators initialized from a zero-mean isotropic distribution $\xi^{(1)} \sim \mathcal{N}(0, I)$, so $\Delta\phi_0 = 0$ while $\Delta\phi_1 > 0$; the downstream budget constraint $K_t \geq 1$ keeps $K = 0$ outside the feasible region.)

(A3) *Direction independence.* The expected projected gain

$$\mathbb{E}\Big[\max_{1 \le j \le K} \langle u, \xi^{(j)} \rangle\Big]$$

does not depend on the unit direction $u \in \mathbb{S}^{d-1}$ (this follows from (A4) for the operators we consider; we list it separately because it is the property used in proofs).

(A4) *Rotational invariance of the joint law.* For any orthogonal matrix $Q \in \mathbb{R}^{d \times d}$, the joint distribution of $\{\xi^{(j)}\}_{1 \le j \le K}$ under $\mathcal{L}$ is invariant under the simultaneous rotation $\xi^{(j)} \mapsto Q\xi^{(j)}$. (For the operators we consider, this follows from the rotational invariance of the initial draw $\xi^{(1)} \sim \mathcal{N}(0, I)$ together with the use of isotropic primitives in subsequent steps; see Remark 14.)

Remark 14 (Grounding of assumptions). Each assumption captures a natural property of practical operators. **(A1)** holds whenever the operator maintains a running best-so-far and each iteration has a positive probability of improving the incumbent. **(A2)** reflects diminishing returns: for random search this follows from concavity of the Gaussian order-statistic constant a_K (David and Nagaraja 2003, Ch. 4); for ϵ-greedy and local perturbation we verify it explicitly below. **(A4)** holds for all three operators we analyze, since each uses only isotropic primitives (Gaussian exploration draws and uniform-direction local perturbations); together with the rotational invariance of the initialization $\xi^{(1)} \sim \mathcal{N}(0, I)$, it implies **(A3)** (direction independence). The factorization $G_t^{\mathcal{L}}(K) = \sigma_t \phi_K^{\mathcal{L}} + O(g_t^2 h)$ is then a *consequence* of (A1)–(A4) plus verifier smoothness (Assumption 4), not an assumption: Theorem 15 below derives it.

C.2 General Gain Factorization

This subsection generalizes the gain decomposition of Theorem 5(iii) from random search to any operator satisfying Assumption 13. The result (Theorem 15) shows that the expected gain of any such operator splits into a timestep-dependent sensitivity factor σ_t and a t-independent operator-specific gain sequence $\phi_K^{\mathcal{L}}$. This separation is what lets the scheduling results in Section 4.3 carry over without change.

Theorem 15 (General gain factorization)**.** *Assume Assumptions 4 and 13 and the iterate moment bound* $\mathbb{E}[\max_{1\le j\le K}\|\xi^{(j)}\|^2] \le M_K(\mathcal{L}, d) < \infty$ *for some sequence* M_K *depending on* K*,* d*, and the operator* $\mathcal{L}$ *(its hyperparameters, e.g., the perturbation radius* λ *for* ϵ*-greedy/local perturbation). Then the expected gain from* K *iterations at timestep* t *satisfies*

$$G_t^{\mathcal{L}}(K) = \sigma_t\,\phi_K^{\mathcal{L}} + O_{K,d,L_t,\mathcal{L}}(g_t^2 h), \tag{19}$$

where $\sigma_t = g_t\sqrt{h}\,\|\nabla\Psi_t(\bar{X}_{t-h}; c)\|$ *and the implicit constant depends on* K*,* d*,* L_t*, and* $M_K(\mathcal{L}, d)$ *(hence on the operator hyperparameters), but not on* g_t *or* $\nabla\Psi_t$*. The properties of* $\phi_K^{\mathcal{L}}$ *(strict monotonicity for* $K \ge 1$*, discrete concavity for* $K \ge 1$*,* $\phi_0^{\mathcal{L}} = 0$*) are exactly the assumptions (A1)/(A2) and the convention.*

Proof. By Proposition 9 applied with $\xi = \xi^{(j)}$ for each iterate (the global Hessian bound in Assumption 4 covers all iterates),

$$\Psi_t(\bar{X}_{t-h} + g_t\sqrt{h}\,\xi; c) = \mu_t + \sigma_t\langle u_t, \xi\rangle + R_t(\xi),$$

where $u_t = \nabla\Psi_t(\bar{X}_{t-h}; c)/\|\nabla\Psi_t(\bar{X}_{t-h}; c)\|$ and $|R_t(\xi)| \le \frac{1}{2}L_t g_t^2 h\|\xi\|^2$. Since μ_t is constant across iterates, substituting into Equation (17) gives

$$\begin{aligned} G_t^{\mathcal{L}}(K) &= \mathbb{E}\Big[\max_{1\le j\le K}\big(\sigma_t\langle u_t, \xi^{(j)}\rangle + R_t(\xi^{(j)})\big)\,\Big|\,X_t, c\Big] \\ &= \sigma_t\,\mathbb{E}\Big[\max_{1\le j\le K}\langle u_t, \xi^{(j)}\rangle\,\Big|\,X_t, c\Big] + O_{K,d,L_t,\mathcal{L}}(g_t^2 h), \end{aligned}$$

where the second equality uses the elementary inequality $|\max_j(a_j + b_j) - \max_j a_j| \le \max_j |b_j|$, the non-negativity $\sigma_t \ge 0$ to factor σ_t out of the maximum, and the moment-bounded remainder

$$\mathbb{E}\big[\max_{1\le j\le K}|R_t(\xi^{(j)})|\big] \le \tfrac{1}{2}L_t g_t^2 h\,\mathbb{E}\big[\max_{1\le j\le K}\|\xi^{(j)}\|^2\big] \le \tfrac{1}{2}L_t g_t^2 h\cdot M_K(\mathcal{L}, d),$$

where the second inequality is the iterate moment bound posited in the theorem statement.

By (A4), the joint law of $\{\xi^{(j)}\}_{1\le j\le K}$ is invariant under simultaneous rotation, so $\mathbb{E}[\max_{1\le j\le K}\langle u_t, \xi^{(j)}\rangle]$ does not depend on u_t (this is the property labeled (A3); see Remark 14). By definition Equation (18), this expectation is $\phi_K^{\mathcal{L}}$. Strict monotonicity in $K \ge 1$ and discrete concavity of $\phi_K^{\mathcal{L}}$ are the assumptions (A1) and (A2) respectively (verified explicitly for each operator in Appendix C.4); $\phi_0^{\mathcal{L}} = 0$ by convention. □

Remark 16 (Sensitivity-gain separation). Theorem 15 establishes the *sensitivity-gain separation*: the timestep-dependent factor σ_t and the operator-dependent gain sequence $\phi_K^{\mathcal{L}}$ decouple multiplicatively. Consequently, the global scheduler needs only the per-step sensitivity profile $\{\sigma_t\}_{t=1}^T$, which is common to all operators, and does not require knowledge of the internal workings of $\mathcal{L}$. Conversely, an operator designer can optimize the gain sequence $\phi_K^{\mathcal{L}}$ independently of the scheduling policy. This modularity underpins the two-level design of GAINS.

C.3 Scheduling Optimality for General Operators

This subsection transfers the offline water-filling result and the online Jensen-gap argument from random search to any operator satisfying Assumption 13. The proofs in Appendix B.3 only invoke three properties of the random-search gain sequence a_K: it is increasing in K, its marginal increments do not accelerate, and it does not depend on the timestep t. Theorem 15 guarantees the same three properties for the operator-specific sequence $\phi_K^{\mathcal{L}}$, so we can replace a_K by $\phi_K^{\mathcal{L}}$ throughout the original proofs.

Corollary 17 (Offline water-filling for general operators)**.** *Under the conditions of Theorem 15 (including the iterate moment bound) with prompt-independent σ_t, and assuming the $O_{K,d,L_t,\mathcal{L}}(g_t^2 h)$ remainder is uniform in (c, x_t) (this is automatic for i.i.d. random search; for non-i.i.d. operators such as ϵ-greedy or local perturbation it amounts to bounding $M_K(\mathcal{L}, d)$ uniformly across initializations), the allocation problem*

$$\max_{\{K_t\}:\, \sum_t K_t = B,\; K_t \geq 1} \sum_{t=1}^{T} \sigma_t\, \phi^{\mathcal{L}}_{K_t} \tag{20}$$

has the same structure as Theorem 6:

(i) *The optimal allocation $\{K_t^*\}$ is simultaneously optimal for every instance.*

(ii) Weak monotonicity*: $\sigma_{t_1} > \sigma_{t_2}$ implies $K^*_{t_1} \geq K^*_{t_2}$. Strictness at the integer level requires the σ-gap to exceed a discrete-marginal step (cf. Theorem 6*(ii)*); the continuous relaxation gives strict monotonicity unconditionally when ϕ_K is strictly concave. When ϕ_K is linear, every optimizer concentrates the allocable budget on $\arg\max_t \sigma_t$ (broken by ties when this set is non-singleton).*

(iii) *Non-uniform allocation weakly dominates uniform; the dominance is strict whenever $\{\sigma_t\}$ are not all equal and either ϕ_K is strictly concave (in the continuous relaxation) or ϕ_K is affine and $B > T$ (the corner solution differs from the uniform interior).*

Proof. The proof splits by curvature. Because Theorem 15 gives a common, t-independent gain sequence $\phi^{\mathcal{L}}_K$ with non-accelerating marginals, the discrete exchange argument from Theorem 6 still shows that an optimal allocation can be chosen monotone in σ_t and that moving budget from a lower-σ_t timestep to a higher-σ_t timestep never hurts, establishing parts (i) and (iii) at the weak level.

When ϕ_K is strictly concave, the KKT argument from Appendix B.3 applies verbatim with ϕ_K in place of a_K, yielding the strict water-filling ordering among active timesteps. When ϕ_K is affine, say $\phi_K = \alpha(K - \beta)$ with $\alpha > 0$ (e.g., local perturbation has $\alpha = c_d(\lambda)$, $\beta = 1$), Equation (20) becomes $\alpha \sum_t \sigma_t K_t - \alpha\beta \sum_t \sigma_t$; the additive constant $-\alpha\beta \sum_t \sigma_t$ does not depend on the allocation, so the problem reduces to a linear program $\alpha \sum_t \sigma_t K_t$, and every optimizer assigns all budget beyond the mandatory floor $K_t \geq 1$ to $\arg\max_t \sigma_t$. □

Remark 18 (Curvature dichotomy in optimal scheduling). The structure of the optimal allocation depends qualitatively on the curvature of $\phi^{\mathcal{L}}_K$. When ϕ_K is strictly concave (e.g., random search, where $\phi_K = a_K \sim \sqrt{2 \log K}$), the optimal schedule distributes budget smoothly across timesteps via water-filling. When ϕ_K is affine in K (e.g., local perturbation in the linearized regime, where $\phi^{(\mathrm{LP})}_K = c_d(\lambda)\,(K-1)$), every optimizer concentrates the allocable budget on $\arg\max_t \sigma_t$. The choice of local search operator therefore influences not only the per-step gain rate but also the *shape* of the optimal global schedule (cf. Table 4).

Corollary 19 (Online Jensen gap and dual-threshold stopping)**.** *Under the conditions of Theorem 15 with prompt-dependent $\sigma_t = \sigma_t(c, x_t)$, define the generalized oracle*

$$V^*_{\mathcal{L}}(\boldsymbol{\sigma}) = \max_{\{K_t\}:\, \sum_t K_t = B,\; K_t \geq 1} \sum_{t=1}^{T} \sigma_t\, \phi^{\mathcal{L}}_{K_t}. \tag{21}$$

Then $V^*_{\mathcal{L}}$ is convex in $\boldsymbol{\sigma}$, and

$$\mathbb{E}_{\boldsymbol{\sigma}}\big[V^*_{\mathcal{L}}(\boldsymbol{\sigma})\big] \geq V^*_{\mathcal{L}}(\bar{\boldsymbol{\sigma}}), \tag{22}$$

with equality whenever the same allocation is optimal almost surely on the support of $\boldsymbol{\sigma}$. The one-step marginal gain satisfies, for integer budgets,

$$\Delta^{\mathcal{L}}_t(K) := G^{\mathcal{L}}_t(K{+}1) - G^{\mathcal{L}}_t(K) = \sigma_t \cdot \Delta\phi^{\mathcal{L}}_K + O_{K,d,L_t,\mathcal{L}}(g_t^2 h), \qquad \Delta\phi^{\mathcal{L}}_K := \phi^{\mathcal{L}}_{K+1} - \phi^{\mathcal{L}}_K. \tag{23}$$

The dual-threshold stopping rule of Algorithm 1 applies as a heuristic proxy control law. For random search the within-step empirical variance $\widehat{\mathrm{Var}}\{\hat{s}^{(j)}_t\}$ consistently estimates σ_t^2 (via Theorem 5(ii)*, since the candidates are i.i.d.); for non-i.i.d. operators (e.g., ϵ-greedy, local perturbation) this estimator is biased because exploit candidates are correlated with the running incumbent, so $\widehat{\mathrm{Var}}\{\hat{s}^{(j)}_t\}$ should be read as a bias-tolerant sensitivity proxy rather than an unbiased estimate of σ_t^2. The empirical effectiveness of this proxy across operators is documented in Appendix D.3.*

Proof. For convexity: for each fixed allocation $\{K_t\}$, the map $\boldsymbol{\sigma} \mapsto \sum_t \sigma_t \, \phi^{\mathcal{L}}_{K_t}$ is linear. $V^*_{\mathcal{L}}$ is the pointwise maximum of finitely many linear functions, hence convex. Jensen's inequality gives Equation (22). This argument uses no property of $\phi^{\mathcal{L}}_K$ beyond it being a real-valued constant for each K.

For the marginal gain: the factorization Equation (23) follows by taking first differences in Equation (19). The dual-threshold logic relies on the product structure of the marginal gain and the non-accelerating property of $\phi^{\mathcal{L}}_K$, both of which persist under (A1)–(A4). The location-scale variance proxy from Theorem 5(ii) is exact for i.i.d. random search and serves as a bias-tolerant heuristic sensitivity proxy for non-i.i.d. operators (as noted in the corollary statement). □

C.4 Operator-Specific Gain Sequences

This subsection instantiates the abstract framework on three concrete local operators: random search, ϵ-greedy search, and local perturbation search (the $\epsilon = 0$ limit of ϵ-greedy). For each operator we derive the gain sequence $\phi^{\mathcal{L}}_K$ in closed form (or as a tight lower bound) and verify the four assumptions (A1)–(A4). All three derivations use the linearized regime $g_t\sqrt{h} \to 0$ in which the verifier reduces to a linear function of the noise, $\Psi_t(\bar{X}_{t-h} + g_t\sqrt{h}\,\xi; c) = \mu_t + \sigma_t\langle u, \xi\rangle + O(g_t^2 h)$ (Proposition 9); the closed-form sequences should therefore be read as the leading-order behavior in this regime.

Example 20 (Random search)**.** The operator draws K i.i.d. $\xi_k \sim \mathcal{N}(0, I)$ and keeps the best. In the linearized regime, the projected scores $\langle u, \xi_k\rangle$ are i.i.d. $\mathcal{N}(0,1)$, so

$$\phi^{\mathrm{RS}}_K = a_K := \mathbb{E}\big[\max(Z_1, \dots, Z_K)\big], \qquad Z_k \overset{\mathrm{iid}}{\sim} \mathcal{N}(0,1). \tag{24}$$

(A1): The running maximum is strictly increasing for $K \geq 2$ (a fresh draw has positive probability of exceeding the current max). **(A2):** For $K \geq 1$, write $\Delta_K = \mathbb{E}[h(M_K)]$ with $h(m) := \mathbb{E}[(Z - m)_+]$ and $M_K := \max(Z_1, \dots, Z_K)$. Since h is strictly decreasing on $\mathbb{R}$ (it is differentiable with $h'(m) = -(1 - \Phi(m)) < 0$) and $M_{K+1} = \max(M_K, Z_{K+1}) \geq M_K$ a.s. with strict inequality on the positive-probability event $\{Z_{K+1} > M_K\}$, we have $h(M_{K+1}) \leq h(M_K)$ a.s. with strict inequality on the same event. Taking expectations gives $\Delta_{K+1} < \Delta_K$ for all $K \geq 1$ (this is the standard concavity of a_K for $K \geq 1$; see David and Nagaraja 2003, Ch. 4). **(A3):** By Theorem 5(iii), $G_t(K) = \sigma_t \, a_K + O_{K,d,L_t}(g_t^2 h)$; the gain sequence $a_K \sim \sqrt{2\log K}$ is t-independent. **(A4):** i.i.d. $\mathcal{N}(0, I)$ draws are rotationally invariant.

Example 21 (ϵ-greedy search)**.** The first iterate is drawn $\xi^{(1)} \sim \mathcal{N}(0, I)$ as the initial incumbent. For $j \geq 2$, with probability ϵ the operator explores by drawing $\xi^{(j)} \sim \mathcal{N}(0, I)$; otherwise it exploits by setting $\xi^{(j)} = \xi^* + RU$ (perturbing the current best ξ^*), where U is uniform on the unit sphere $\mathbb{S}^{d-1}$ and $R \sim \mathrm{Unif}[0, \lambda\sqrt{d}]$, and updates ξ^* if the new candidate is better. In the linearized regime, the verifier satisfies $\Psi_t = \mu_t + \sigma_t\langle u, \xi\rangle + O(g_t^2 h)$ (Proposition 9), so the operator's accept/reject test on Ψ_t coincides with the projected test on $\langle u, \xi\rangle$ to leading order; in particular, the incumbent ξ^* also realizes the projected best-so-far $\langle u, \xi^*\rangle = \max_{j \leq K}\langle u, \xi^{(j)}\rangle$. Then:

- *Exploration steps* (random count $N + 1 \sim 1 + \mathrm{Bin}(K-1, \epsilon)$): each is an i.i.d. draw from $\mathcal{N}(0,1)$ in the projected coordinate $\langle u, \xi\rangle$, contributing a gain of $\mathbb{E}[a_{N+1}]$.
- *Exploitation steps* (expected count $(1-\epsilon)(K-1)$, since $\xi^{(1)}$ is always an exploration draw): each perturbs the current best in the projected direction by $R\langle u, U\rangle$. The accepted increment is $(R\langle u, U\rangle)_+$, whose expectation is the uniform-ball drift constant

$$c_d(\lambda) := \mathbb{E}\big[(R\langle u, U\rangle)_+\big] = \frac{\lambda\sqrt{d}}{4\sqrt{\pi}}\,\frac{\Gamma(d/2)}{\Gamma((d+1)/2)}. \tag{25}$$

Thus exploitation adds an affine-in-K drift beyond the exploration component. Let $N \sim \mathrm{Bin}(K-1, \epsilon)$ count the number of exploration steps among $\xi^{(2)}, \dots, \xi^{(K)}$ (the initial $\xi^{(1)} \sim \mathcal{N}(0, I)$ is always an exploration draw). Conditional on N, the exploration contribution to the projected best is a_{N+1} in expectation, so we retain the lower bound

$$\phi^{(\epsilon)}_K \geq \mathbb{E}\big[a_{N+1}\big], \qquad N \sim \mathrm{Bin}(K-1, \epsilon), \tag{26}$$

with equality when exploitation steps add no improvement (e.g., $\lambda \to 0$). When $\epsilon = 1$, $N = K - 1$ deterministically and the operator reduces to random search with $\phi^{(1)}_K = a_K$.

(A1): For $K \geq 1$, each subsequent iteration has probability $\epsilon > 0$ of being an exploration step that draws a fresh $Z_{\text{new}} \sim \mathcal{N}(0,1)$ in the projected coordinate. Because $\mathcal{N}(0,1)$ has unbounded support, $P(Z_{\text{new}} > \max_{1\leq j\leq K}\langle u, \xi^{(j)}\rangle) > 0$, so $\phi^{(\epsilon)}_{K+1} > \phi^{(\epsilon)}_K$ for all $K \geq 1$ (matching the $K \geq 1$ scope of (A1); at $K = 0$, $\phi_1 = \phi_0 = 0$ since $\xi^{(1)}$ has zero-mean projection).

(A2): For $K \geq 1$, let $M_K = \max_{1\leq j\leq K}\langle u, \xi^{(j)}\rangle$ and $h(m) := \mathbb{E}[(Z-m)_+]$ for $Z \sim \mathcal{N}(0,1)$ (strictly decreasing). The marginal gain decomposes as

$$\Delta_K = \underbrace{\epsilon \cdot \mathbb{E}[h(M_K)]}_{\text{explore}} + \underbrace{(1-\epsilon)\, c_d(\lambda)}_{\text{exploit}},$$

since on an exploration step $\langle u, \xi^{(K+1)}\rangle \sim \mathcal{N}(0,1)$ is independent of M_K, and on an exploitation step the projected increment $(R\langle u, U\rangle)_+$ is independent of M_K with expectation $c_d(\lambda)$. The exploit term is constant in K. For the explore term: $M_{K+1} \geq M_K$ *a.s.* (running max under either step type), so $h(M_{K+1}) \leq h(M_K)$ a.s.; on the positive-probability event $\{M_{K+1} > M_K\}$ (which occurs whenever step $K+1$ improves the incumbent, e.g., probability $\epsilon \cdot \Pr(Z > M_K) + (1-\epsilon)\cdot \frac{1}{2} > 0$), the inequality is strict. Taking expectations: $\mathbb{E}[h(M_{K+1})] < \mathbb{E}[h(M_K)]$ for all $K \geq 1$. Hence $\Delta_{K+1} - \Delta_K = \epsilon \cdot (\mathbb{E}[h(M_{K+1})] - \mathbb{E}[h(M_K)]) < 0$ when $\epsilon > 0$ (and equals 0 when $\epsilon = 0$, which reduces to Example 22); (A2) holds for $K \geq 1$, strictly when $\epsilon > 0$.

(A3): In the linearized regime, both the exploration draw $\langle u, \xi_{\text{new}}\rangle \sim \mathcal{N}(0,1)$ and the exploitation perturbation $R\langle u, U\rangle$ depend on u only through a one-dimensional projection, whose distribution is u-independent by isotropy. Hence $\phi^{(\epsilon)}_K$ does not depend on the timestep t.

(A4): Both primitives (Gaussian draws for explore, sphere-uniform directions with scalar radii for exploit) are rotationally invariant; the accept-reject comparison uses $\langle u, \xi\rangle$, which satisfies $\langle Qu, Q\xi\rangle = \langle u, \xi\rangle$ for any orthogonal Q.

Example 22 (Local perturbation search (ϵ-greedy with $\epsilon = 0$))**.** This operator is the pure-exploitation limit of Example 21: each iteration perturbs the current best ξ^* by RU with $U \sim \text{Unif}(\mathbb{S}^{d-1})$ and $R \sim \text{Unif}[0, \lambda\sqrt{d}]$, setting $\xi_{\text{new}} = \xi^* + RU$, and updates ξ^* if the new candidate scores higher. In the linearized regime, the projected coordinate of the incumbent (which equals the projected best-so-far, since the verifier is linear in ξ to leading order) evolves by accept-reject: the perturbation $R\langle u, U\rangle$ is accepted when $\langle u, \xi_{\text{new}}\rangle > \langle u, \xi^*\rangle$, i.e., when $\langle u, U\rangle > 0$. Because the acceptance event is independent of the current incumbent, each iteration $j \geq 2$ adds a constant expected improvement, giving the marginal recursion

$$\phi^{(\text{LP})}_{K+1} - \phi^{(\text{LP})}_K = c_d(\lambda) \quad (K \geq 1), \tag{27}$$

and, combined with the convention $\phi^{(\text{LP})}_0 = \phi^{(\text{LP})}_1 = 0$ (the initial iterate $\xi^{(1)} \sim \mathcal{N}(0, I)$ has zero-mean projection),

$$\phi^{(\text{LP})}_K = c_d(\lambda)\,(K-1) \quad (K \geq 1). \tag{28}$$

(A1): Each perturbation has probability $P(\langle u, U\rangle > 0) = 1/2 > 0$ of improving the incumbent, so ϕ_K is strictly increasing for $K \geq 1$. **(A2):** The marginal gain Equation (27) is the constant $c_d(\lambda)$ for all $K \geq 1$, so $\Delta_{K+1} = \Delta_K$ trivially satisfies the non-increasing condition (with equality). The resulting $\phi^{(\text{LP})}_K = c_d(\lambda)\,(K-1)$ is affine in K (weakly concave but not strictly), which is why Corollary 17(ii) treats this case separately: under affine ϕ_K, every optimizer concentrates allocable budget on $\arg\max_t \sigma_t$ rather than spreading it via water-filling. Outside the linearized regime, curvature effects perturb this exact constant-marginal picture; the abstract theory uses Equation (28) only as the leading-order gain sequence. **(A3):** The perturbation $R\langle u, U\rangle$ is a one-dimensional projection of a uniform-ball perturbation, whose distribution does not depend on u or t. Hence $\phi^{(\text{LP})}_K$ is t-independent. **(A4):** Same as the ϵ-greedy case.

Table 4 shows that the three operators differ qualitatively in growth rate. Random search saturates logarithmically, while local perturbation grows linearly in the linearized regime. This implies that the optimal operator choice depends on the per-step budget: smaller budgets are well served by random search, while at larger budgets the linear-growth local perturbation operator becomes relatively more attractive. A precise crossover analysis would require characterizing how the higher-order remainder $R_t(\xi) = O(g_t^2 h\|\xi\|^2)$ couples to the operator-dependent iterate norms outside the linearized regime, where the iterates of local perturbation may accumulate drift; we leave this to future work and treat the within-regime ordering as a guideline rather than a theorem.

Table 4: Summary of operator-specific gain sequences $\phi_K^{\mathcal{L}}$ in the linearized regime. All operators satisfy (A1)–(A4). The growth rate determines how quickly search saturates; faster-growing operators benefit more from online scheduling (Remark 18).

Operator $\mathcal{L}$	$\phi_K^{\mathcal{L}}$ (linearized)	Growth rate	Regime note
Random search	$a_K \sim \sqrt{2 \log K}$	$O(\sqrt{\log K})$	Exact
ϵ-greedy	$\geq \mathbb{E}[a_{N+1}]$, $N{\sim}\mathrm{Bin}(K{-}1, \epsilon)$	$\Omega(\sqrt{\log K})$	Lower bound
Local perturbation (ϵ=0)	$c_d(\lambda)\,(K-1)$	$O(K)$	Exact (linearized, uniform-ball)

D Additional Experiments and Detailed Analysis

This appendix collects four supporting experiments referenced in Section 5 and reads each one against the theory. The four are: (i) an ablation that separates the offline allocation stage of GAINS from its online control stage (Appendix D.1); (ii) a generalization study on a larger and more diverse prompt set (Appendix D.2); (iii) a compatibility test that swaps the local search operator while keeping the global scheduler fixed (Appendix D.3); and (iv) an evaluation on flow-based models, where stochasticity is reduced compared to standard diffusion (Appendix D.4). Each subsection reports the corresponding table and discusses what the numbers reveal about the scheduling mechanism predicted by the theory.

D.1 Ablation: Offline vs. Offline+Online

This subsection isolates the two scheduling stages of GAINS (offline allocation across timesteps and online per-prompt control) and measures how much each stage contributes. Table 5 reports Stable Diffusion results at a fixed total budget of 400 NFE. The uniform-allocation benchmark achieves Brightness 0.7025 and Compressibility 0.8833. Replacing the uniform allocation with the offline schedule alone (the per-timestep budgets $\{K_t\}$ are fixed in advance, and online early stopping is disabled) already lifts both metrics by $+0.0133$ on Brightness and $+0.0040$ on Compressibility, indicating that even coarse per-timestep budgeting captures genuine model-level structure that does not require any per-prompt adjustment. Adding online feedback on top (the full GAINS) yields the highest scores, 0.7248 Brightness and 0.8946 Compressibility, an incremental $+0.0090$ on Brightness and $+0.0073$ on Compressibility beyond the offline-only setting.

Table 5: Stable Diffusion at fixed NFE $= 400$. **Uniform**: uniform per-step allocation. **GAINS**: offline profiling plus online control. **Offline Only**: offline allocation without online control. Online control contributes incremental gains beyond offline allocation alone on both metrics.

Metric	**Uniform**	**GAINS**	**Offline Only**
Brightness	0.7025 ± 0.047	$\mathbf{0.7248 \pm 0.056}$	0.7158 ± 0.065
Compressibility	0.8833 ± 0.017	$\mathbf{0.8946 \pm 0.016}$	0.8873 ± 0.018

This decomposition matches the two-factor structure predicted by Theorem 5(iii) and Remark 8: offline profiling identifies *where* search matters by estimating the population-mean sensitivity profile $\bar{\boldsymbol{\sigma}}$, while online control then refines *how much* budget is worth spending on the specific prompt at hand by reading the per-instance realization $\boldsymbol{\sigma}(c, x_t)$. The incremental gain from online over offline-only ($+0.0090$ Brightness, $+0.0073$ Compressibility) is the empirical Jensen gap $\mathbb{E}_{\boldsymbol{\sigma}}[V^*(\boldsymbol{\sigma})] - V^*(\bar{\boldsymbol{\sigma}})$: because per-prompt sensitivities vary, a single allocation tuned to the mean profile leaves value on the table, and the online controller recovers part of that gap.

D.2 Larger Prompt Set: Generalization

This subsection tests whether the offline schedule transfers to a broader and more diverse prompt distribution. The natural concern about offline profiling is overfitting to the calibration set: if the profiled sensitivities $\hat{\boldsymbol{\sigma}}$ reflect quirks of the specific calibration prompts rather than population-level structure, the gains observed on the calibration prompts will not transfer to held-out prompts. To probe this, we expand the evaluation to 20 prompts with 10 independent samples each (Table 6).

Table 6: Larger prompt set evaluation: 20 prompts, each repeated 10 times. Improvements persist across a diverse prompt set, confirming systematic reallocation rather than prompt-specific artifacts.

	Brightness	**Compressibility**
Uniform	0.6949 ± 0.076	0.7938 ± 0.089
GAINS	$\mathbf{0.7213 \pm 0.075}$	$\mathbf{0.8076 \pm 0.096}$

GAINS raises mean Brightness from 0.6949 to 0.7213 (a gain of $+0.0264$) and mean Compressibility from 0.7938 to 0.8076 (a gain of $+0.0138$). The standard deviations remain comparable across the two methods, so the improvement is not driven by selectively boosting a few easy prompts. Since the gains persist under repeated sampling across this more diverse prompt set, the benefit reflects systematic reallocation of compute toward high-impact timesteps rather than cherry-picked prompt selection. This generalization matches Theorem 6(i): when σ_t depends primarily on the timestep index rather than on the specific prompt, a single offline allocation is simultaneously near-optimal for every prompt in the population.

D.3 Compatibility with Local Search Operators

This subsection tests the modularity prediction of Corollary 17: since global scheduling depends only on the sensitivity profile $\{\sigma_t\}$ and not on the operator-specific gain sequence $\phi_K^{\mathcal{L}}$, GAINS should improve over uniform allocation regardless of which local search operator is used inside each timestep, provided the operator satisfies (A1)–(A4). Table 7 runs both the uniform benchmark and GAINS under two qualitatively different operators: **Zero-order** (local perturbation, Example 22) and **Random** search (Example 20).

Table 7: Global scheduling combined with different local search operators. Left block uses uniform allocation; right block uses GAINS. GAINS improves both metrics under every local operator, confirming that global scheduling composes with the local search mechanism.

	Uniform		**GAINS**	
Metric	**Zero-order**	**Random**	**Zero-order**	**Random**
B (Brightness)	0.4920 ± 0.072	0.7382 ± 0.039	$\mathbf{0.5080 \pm 0.075}$	$\mathbf{0.7457 \pm 0.045}$
C (Compressibility)	0.5727 ± 0.066	0.8417 ± 0.027	$\mathbf{0.5832 \pm 0.061}$	$\mathbf{0.8454 \pm 0.025}$

The uniform-allocation baselines differ sharply between the two operators (Zero-order Brightness 0.4920 vs. Random Brightness 0.7382), reflecting the curvature dichotomy in Remark 18: random search yields a broad logarithmic gain $a_K \sim \sqrt{2 \log K}$, while pure local perturbation yields an affine gain $c_d(\lambda)(K-1)$ that is highly sensitive to the perturbation radius λ. Despite this large baseline gap, GAINS improves *both* operators on *both* metrics: $+0.0160$ on Brightness and $+0.0105$ on Compressibility for Zero-order, and $+0.0075$ on Brightness and $+0.0037$ on Compressibility for Random. The global scheduler targets where to spend compute and pairs with different per-timestep proposal mechanisms without modification, exactly as predicted by the sensitivity-gain separation (Remark 16).

D.4 Flow-Based Models: Reduced Stochasticity

This subsection extends the evaluation to flow-based generative models, where the per-step transition is deterministic and so contains no noise variable to search over. Following Singh and Fischer [2024] and Kim et al. [2025], we inject Gaussian stochasticity at a small set of selected timesteps, converting the original ordinary differential equation (ODE) sampler into a stochastic differential equation (SDE) sampler with matching marginal distributions. Once written in the per-step form of Equation (2), the resulting sampler admits the same noise-trajectory search as standard diffusion, and GAINS applies without modifying the pretrained flow model.

Global scheduling improves quality even under the reduced stochasticity of flow-based models. For Brightness, GAINS yields gains of $+0.0015$, $+0.0020$, and $+0.0011$ at 80, 100, and 150 NFE; the 80-NFE result (0.5507) already matches and slightly exceeds uniform allocation at 100 NFE

Table 8: Flow-based model results. Stochasticity is introduced via noise injection [Singh and Fischer, 2024], after which GAINS allocates search compute non-uniformly across timesteps. Margins are smaller than on SD and EDM, consistent with the narrower variance dispersion after ODE-to-SDE conversion.

NFE	Uniform	GAINS
	Brightness	
80	0.5492 ± 0.003	$\mathbf{0.5507 \pm 0.003}$
100	0.5499 ± 0.003	$\mathbf{0.5519 \pm 0.003}$
150	0.5515 ± 0.004	$\mathbf{0.5526 \pm 0.003}$
	Compressibility	
80	0.8135 ± 0.005	$\mathbf{0.8162 \pm 0.003}$
100	0.8156 ± 0.004	$\mathbf{0.8170 \pm 0.006}$
150	0.8157 ± 0.006	$\mathbf{0.8193 \pm 0.005}$

(0.5499), saving **20%** of the compute budget. For Compressibility, the improvements reach $+0.0027$, $+0.0014$, and $+0.0036$; the 100-NFE GAINS score (0.8170) surpasses uniform allocation at 150 NFE (0.8157), a **33%** reduction in compute.

The smaller absolute margins compared to Stable Diffusion and EDM (where the gap reaches $+0.03$ to $+0.04$) match the prediction of Theorem 6(iii): the advantage of non-uniform scheduling over uniform scheduling grows with the dispersion of the sensitivity profile $\{\sigma_t\}$ across active timesteps. The ODE-to-SDE conversion injects noise only at a few selected timesteps; restricted to those active steps, the resulting profile is more concentrated and less heterogeneous than in fully stochastic models, which shrinks the available reallocation benefit. Even so, the quality curve shifts to the left at every budget, and GAINS also reduces the run-to-run standard deviations (for example, from 0.0058 to 0.0046 for Compressibility at 150 NFE), indicating that even modest reallocation tightens the per-prompt distribution of outcomes.

NeurIPS Paper Checklist

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   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction state three core contributions: (i) the two-level framework for noise trajectory search (Section 3); (ii) the GAINS algorithm combining offline profiling with online control (Section 4); (iii) theoretical foundations including the location-scale gain factorization (Theorem 5), offline water-filling (Theorem 6), and an $\Omega(T)$ regret lower bound (Lemma 7). All claims are supported by the formal results in Section 4.3 and the experiments in Section 5; the claimed 20–50% NFE reduction matches the numbers reported in Tables 1, 2 and 8.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: The "Limitations and future work" paragraph in Section 6 explicitly discusses three limitations: the light-tailed location-scale assumption, prompt-stationarity of profiled sensitivities, and the use of two lightweight verifiers in our experiments.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

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   Justification: Assumption 4 (verifier smoothness) and Assumption 13 (local-search regularity) are stated formally before the theorems that depend on them. Complete proofs of Theorem 5, Theorem 6, and Lemma 7 appear in Appendix B; the full proof of the general factorization Theorem 15 appears in Appendix C.

4. **Experimental result reproducibility**

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7. **Experiment statistical significance**

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   Answer: [Yes]

   Justification: Section 5 states that all experiments run on a single NVIDIA A100 GPU and that wall-clock time is dominated by forward passes through the denoising network and is therefore determined by the prescribed NFE budget.

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10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [N/A]

    Justification: The contribution is an inference-time scheduling method that reduces compute consumption for existing pretrained diffusion models without changing their generative capabilities. It does not enable new misuse pathways beyond those already present in the underlying models (e.g., Stable Diffusion, EDM); positive impact is reduced compute and energy use during deployment.

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pretrained language models, image generators, or scraped datasets)?

    Answer: [N/A]

    Justification: The paper does not release new datasets or pretrained generative models. We propose an inference-time scheduling algorithm that operates on existing pretrained models without modification.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: [Yes]

Justification: All pretrained models used (Stable Diffusion, EDM, flow-matching models) are properly cited in Sections 1.1 and 5. The reference implementation we build on [Ramesh and Mardani, 2025] is properly cited as well. We use these assets in accordance with their respective licenses (CreativeML Open RAIL-M for Stable Diffusion; CC BY-NC-SA 4.0 for EDM).

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [N/A]

    Justification: The paper does not introduce new datasets or pretrained models. The code implementing GAINS will be released upon acceptance with documentation of all hyperparameters and usage instructions.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: [N/A]

    Justification: The paper does not involve crowdsourcing or human-subject research. All experiments use deterministic verifiers (Brightness, Compressibility) computed directly from generated images.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: [N/A]

    Justification: The paper does not involve human subjects.

16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research?

    Answer: [N/A]

    Justification: The proposed method, GAINS, is an inference-time scheduling algorithm for diffusion models and does not use large language models as a component of the methodology. LLMs were used only for routine writing assistance (e.g., grammar polishing), which the NeurIPS LLM policy explicitly exempts from disclosure.